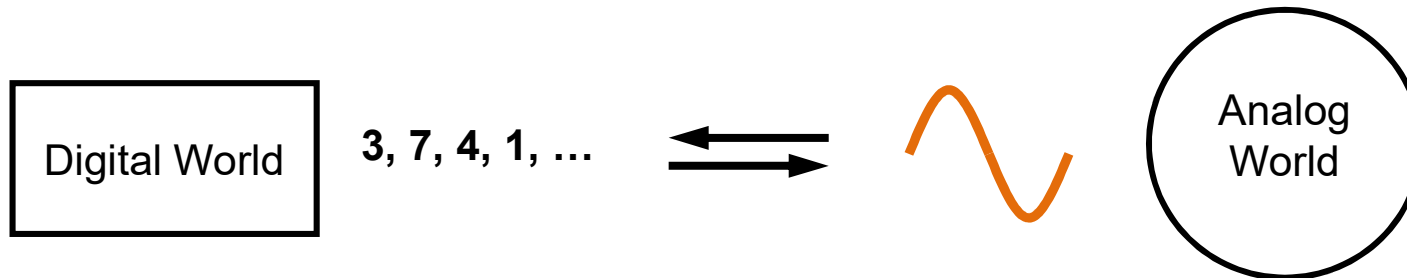


Data Converter High-Level Picture

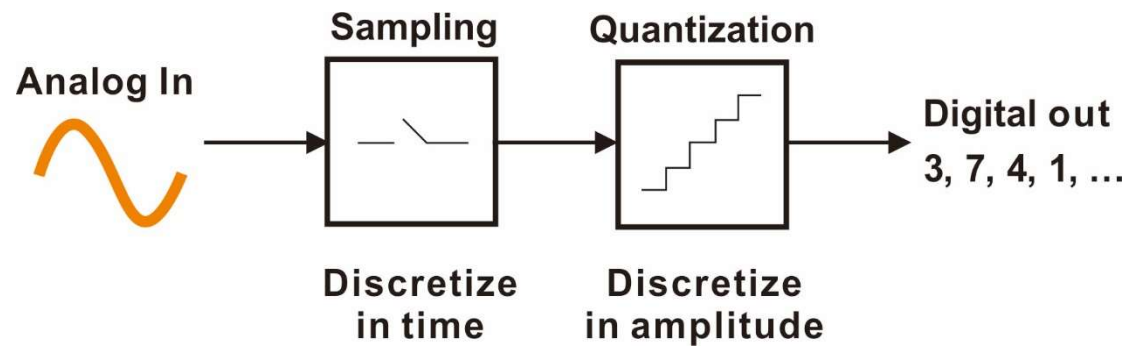
Nan Sun
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Data Conversion

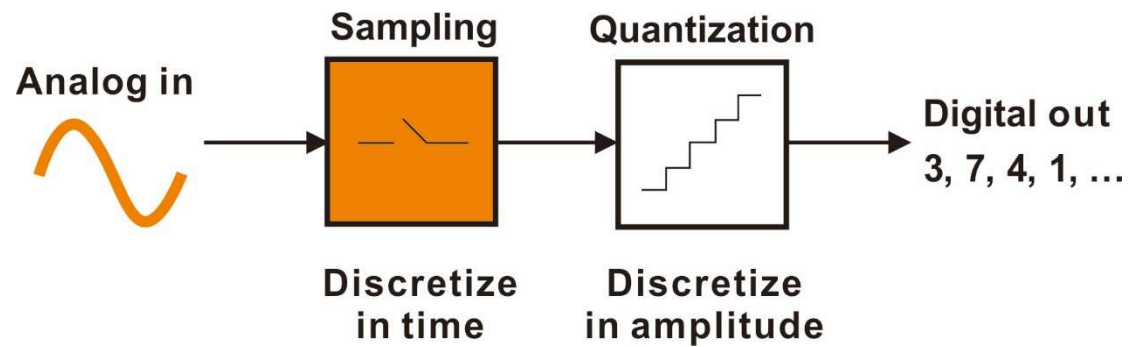


- Analog signal: $a(t)$
 - Continuous time, continuous amplitude
- Digital signal: $d[n]$
 - Discrete time, discrete amplitude
- $a(t) \rightarrow d[n]$
 - discretize in time and amplitude
- $d[n] \rightarrow a(t)$
 - “continuize” in time and amplitude

Analog-to-Digital Conversion



Analog-to-Digital Conversion



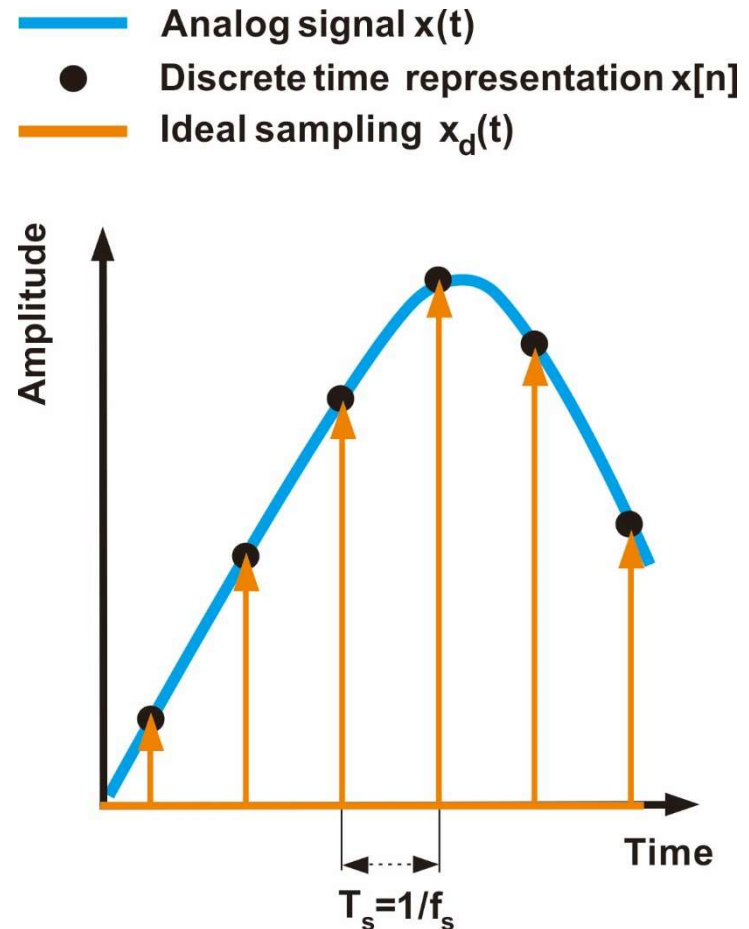
Ideal Sampling

- Ideal sampling

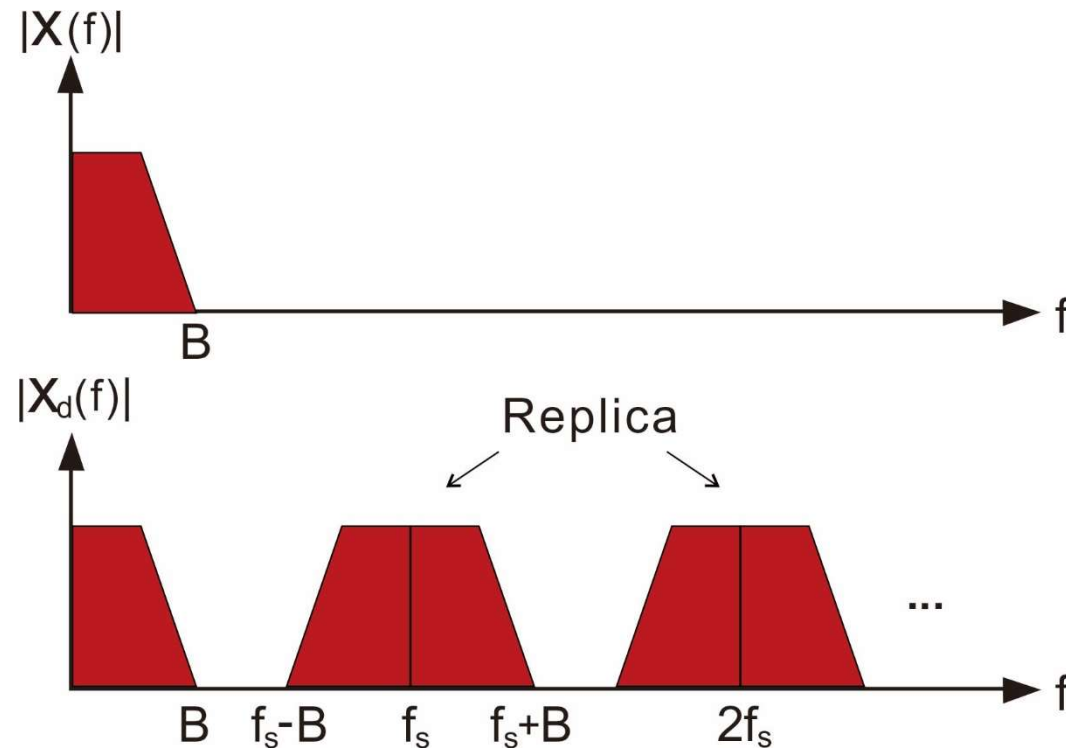
$$x_d(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

- Multiplication in time means convolution in frequency

$$X_d(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s)$$

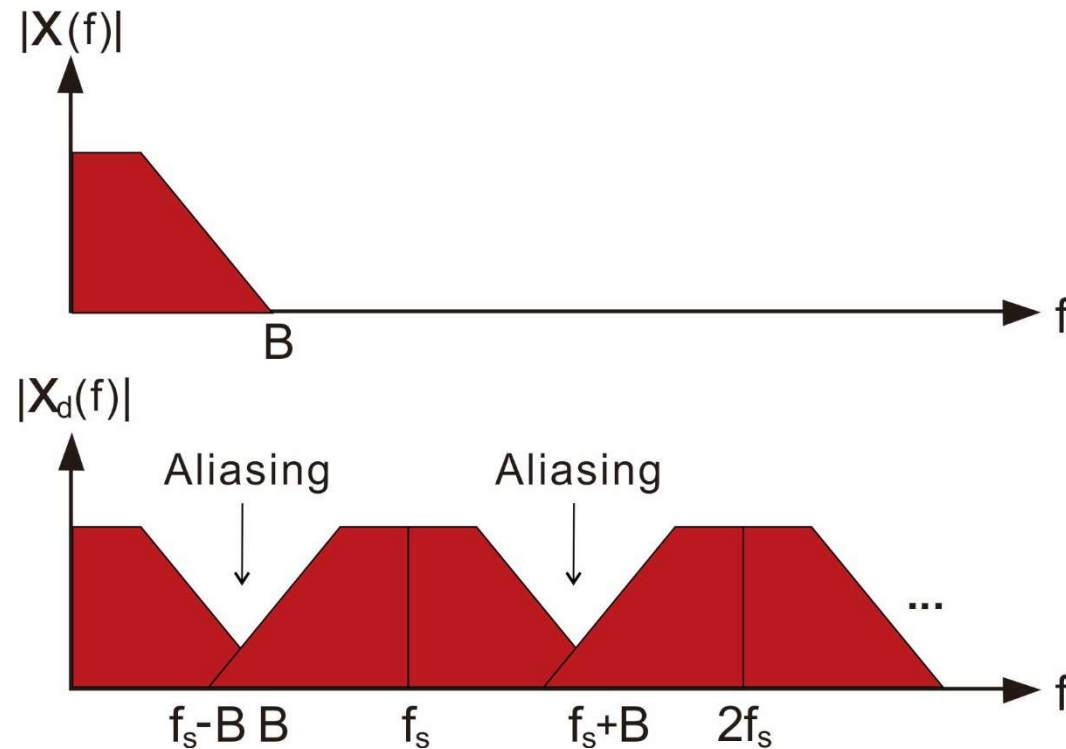


No Aliasing



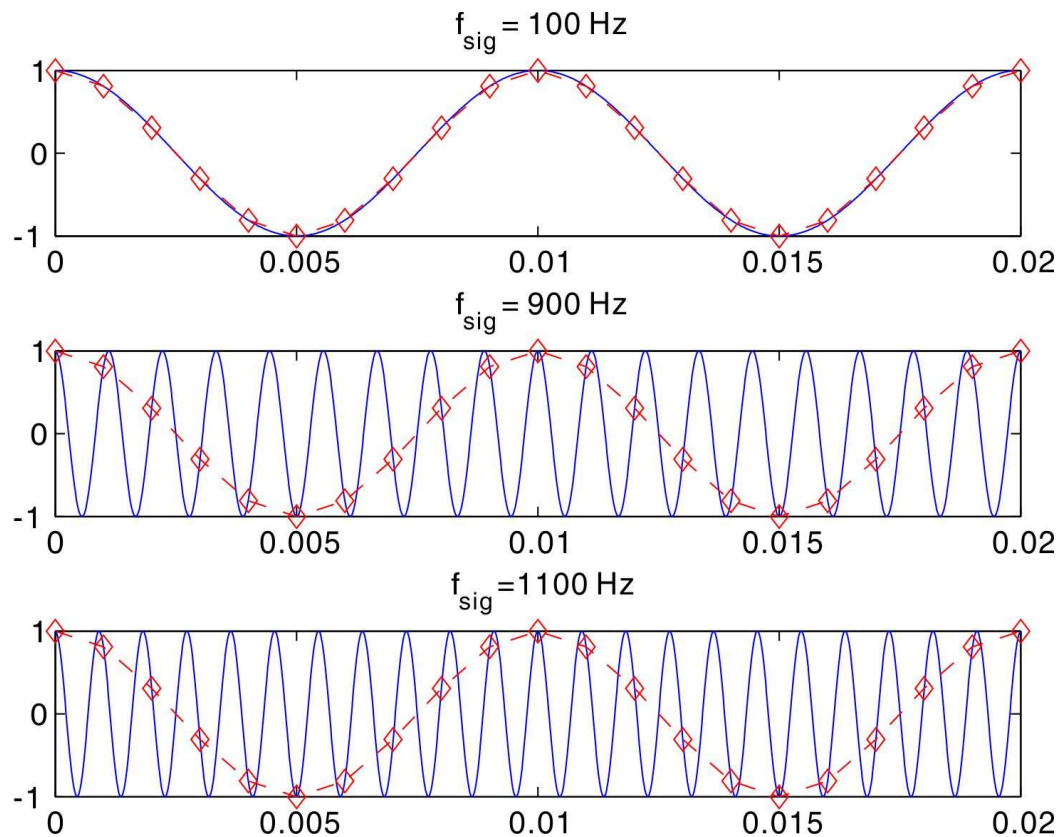
- $X_d(f)$ contains replicas of $X(f)$ at Nf_s
- No aliasing for $f_s - B > B \rightarrow f_s > 2B$

Aliasing



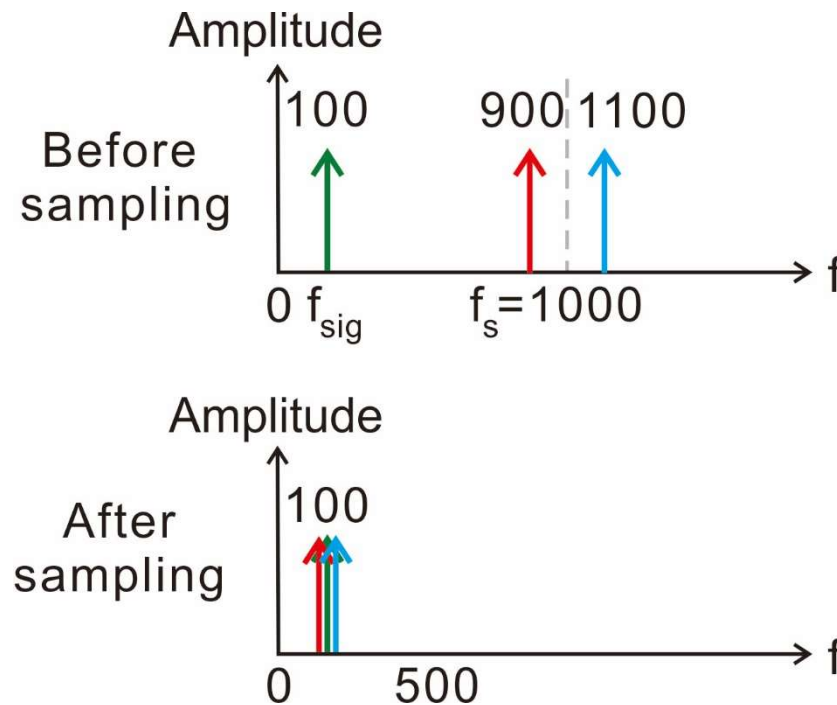
- Aliasing for $f_s < 2B$
- We cannot distinguish signals with different frequencies as they overlap on top of each other after sampling.

Aliasing Example



- $f_s = 1000$ Hz
- $V_{\text{in}} = \cos(2\pi f_{\text{sig}} t)$
 $V_d = \cos[2\pi(f_{\text{sig}} n T_s)]$
 $= \cos[2\pi(f_{\text{sig}}/f_s)n]$
- Signals with different frequencies yield the same sampled result.

Frequency-Domain View

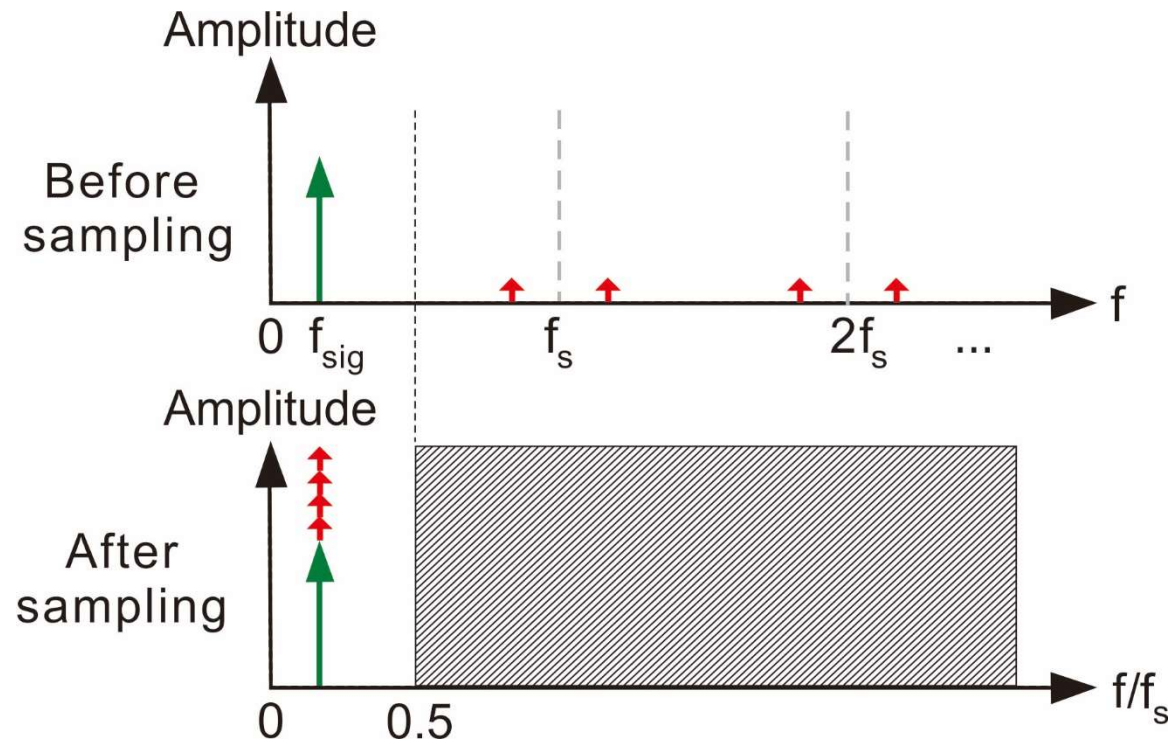


- Components at $N \cdot f_s \pm f_{sig}$ (N integer) are indistinguishable after sampling

Nyquist Sampling Theorem

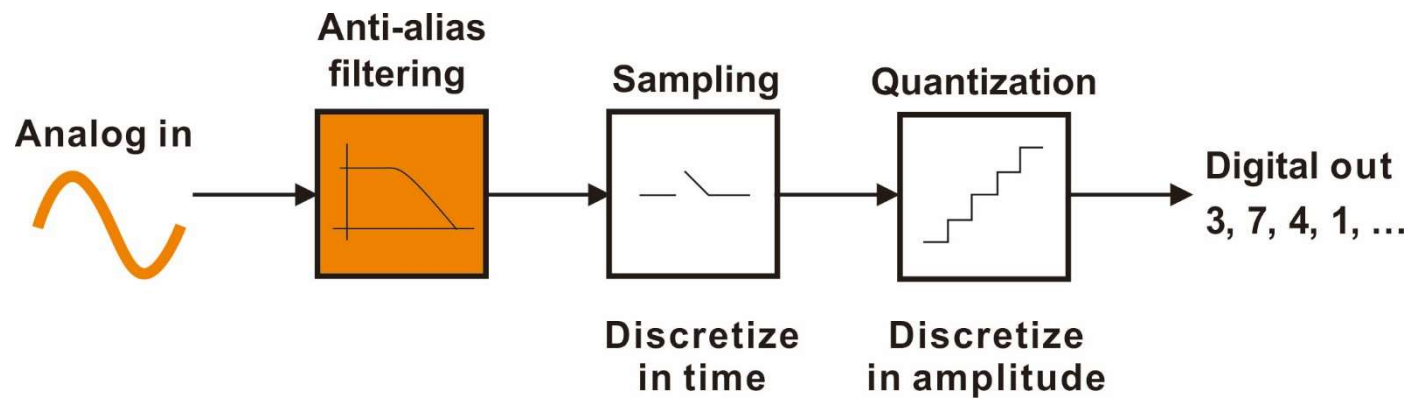
- In order to prevent aliasing, we need $f_s > 2B$.
- The sampling rate $f_s = 2B$ is called the **Nyquist rate**.
- Question: what if there are out-of-band interferes and noise?
 - Increase B to contain most interferes and noise
 - Limit B through filtering (signal conditioning)

Without Anti-Aliasing Filter

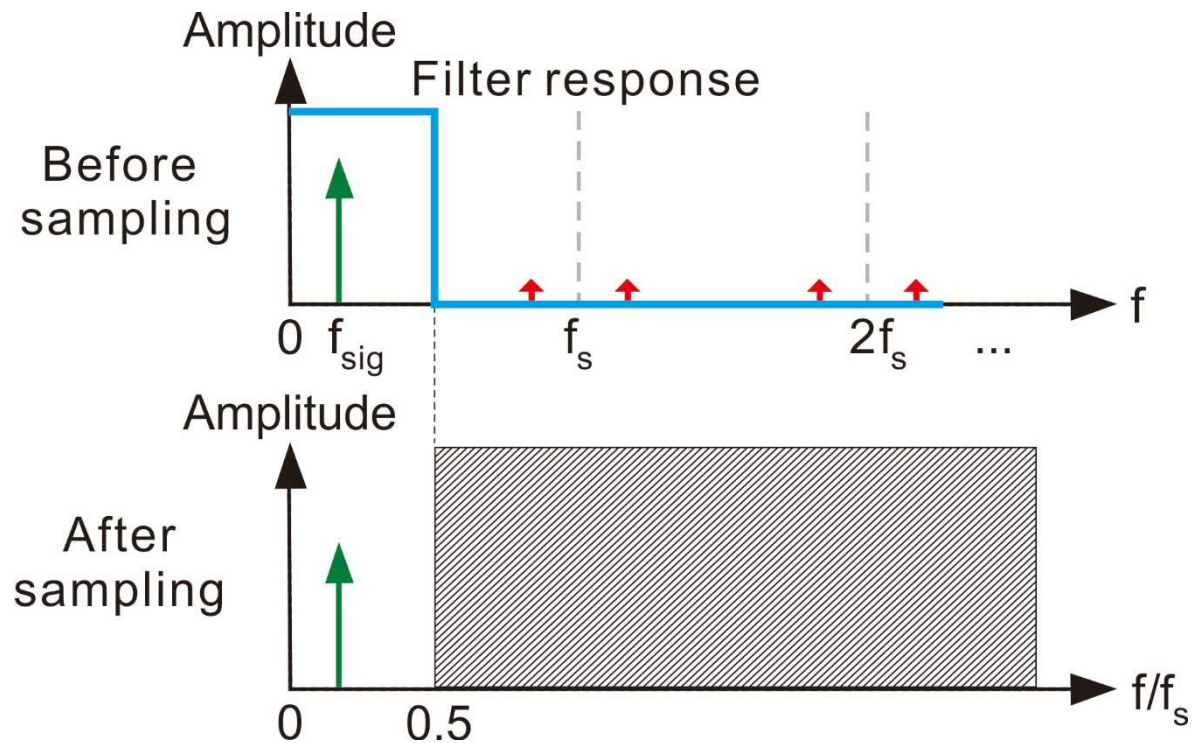


- Unwanted interferers and noise are folded into the signal band.

Analog-to-Digital Conversion

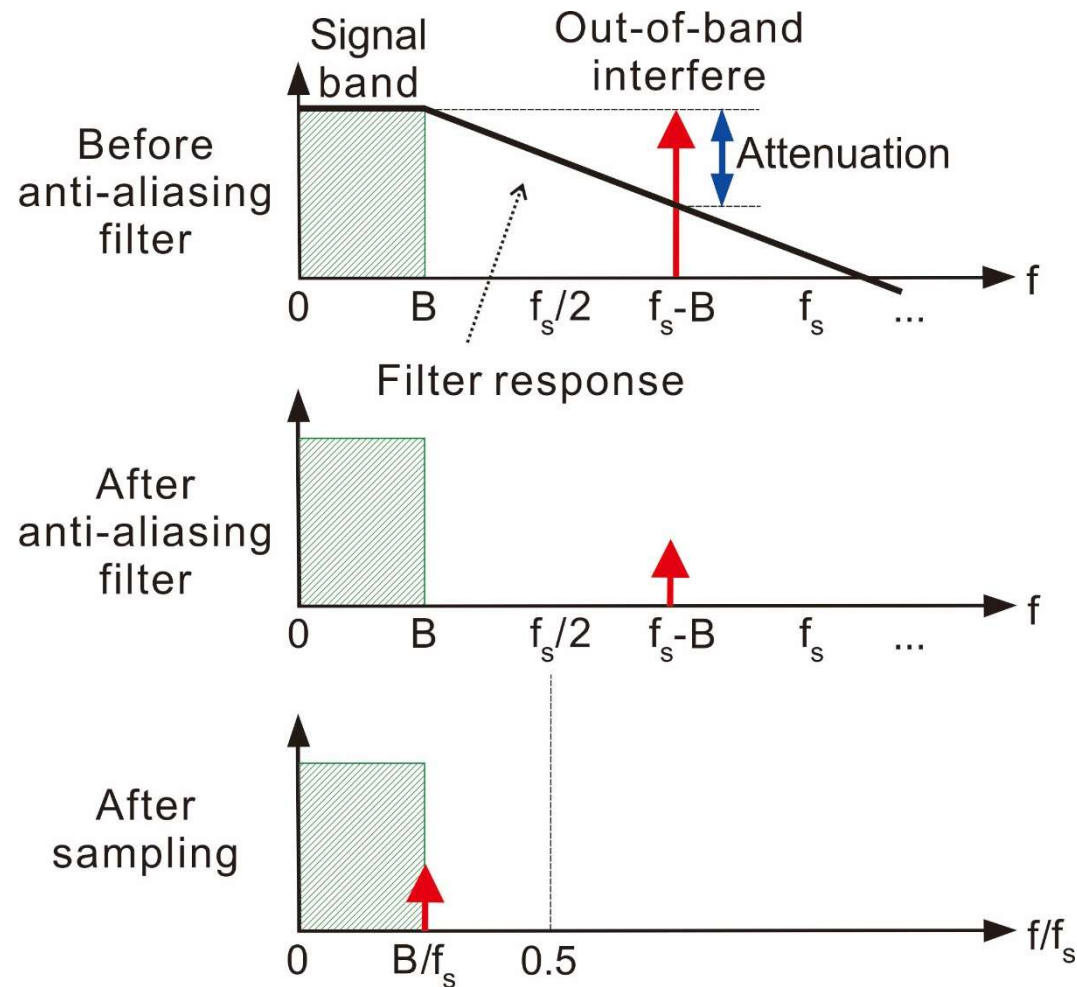


Brick Wall Anti-Aliasing Filter



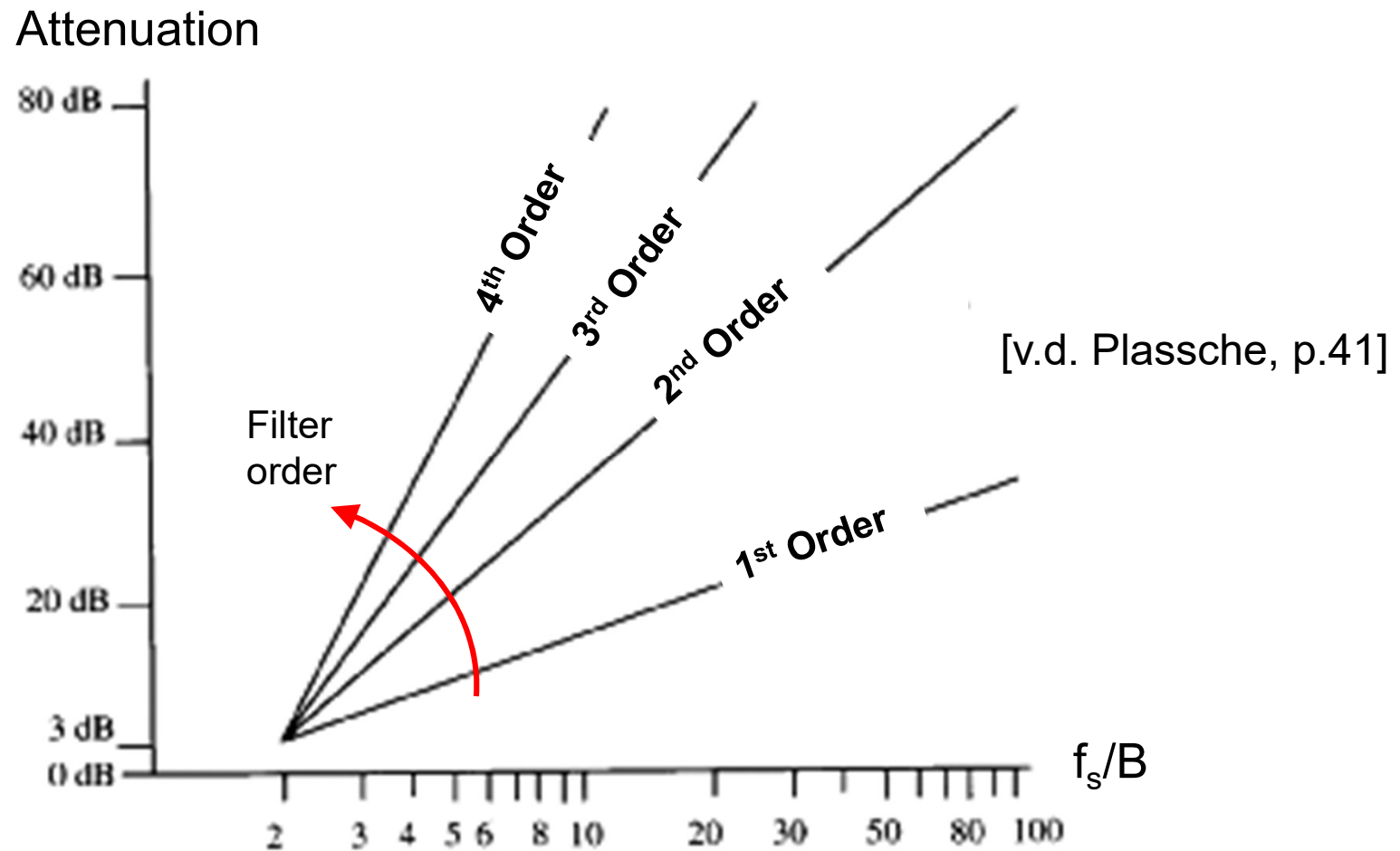
- Use an analog filter to reject unwanted signals.

Practical Anti-Aliasing Filter



- How to increase attenuation? 1) increase f_s/B ; 2) increase filter order

Attenuation vs. Oversampling & Filter Order

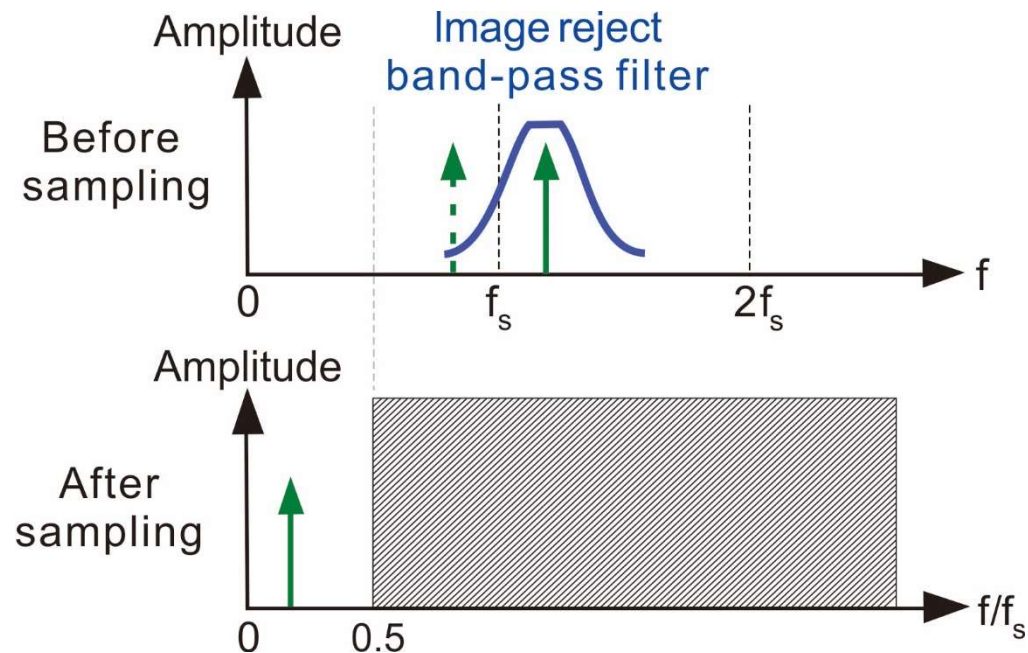


We can tradeoff oversampling ratio with filter order

Classes of Sampling

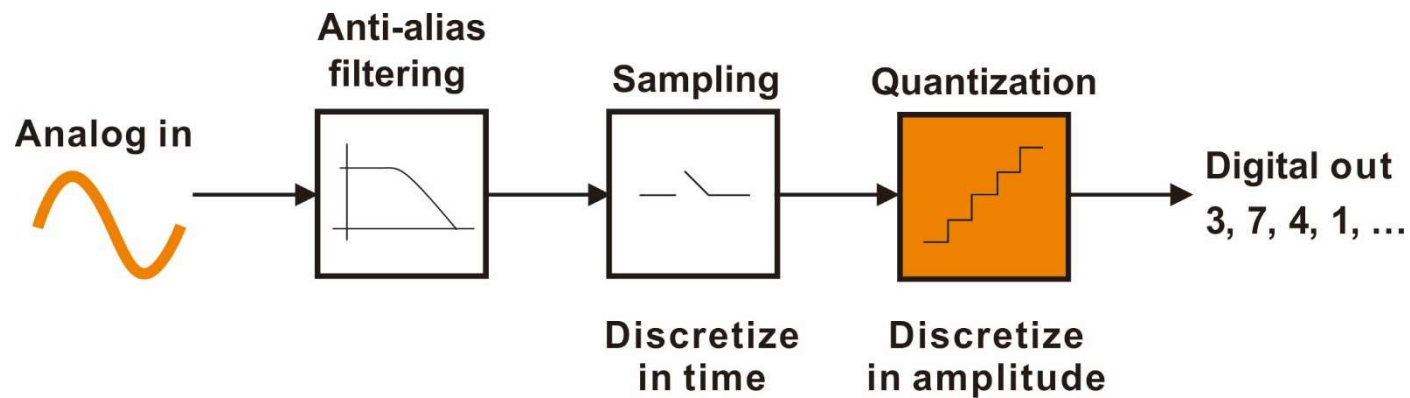
- Nyquist-rate sampling ($f_s > 2B$)
 - Nyquist-rate data converters
 - In practice always slightly oversampled (e.g., by 2 or 4)
- Oversampling ($f_s \gg 2B$)
 - Oversampled data converters
 - Anti-alias filtering is often trivial
 - Oversampling also reduces quantization noise
 - More details to follow
- Subsampling ($f_s < 2B$)
 - Exploit aliasing to down-convert high-frequency signals
 - See [Pekau & Haslett, JSSC, 2005]

Subsampling

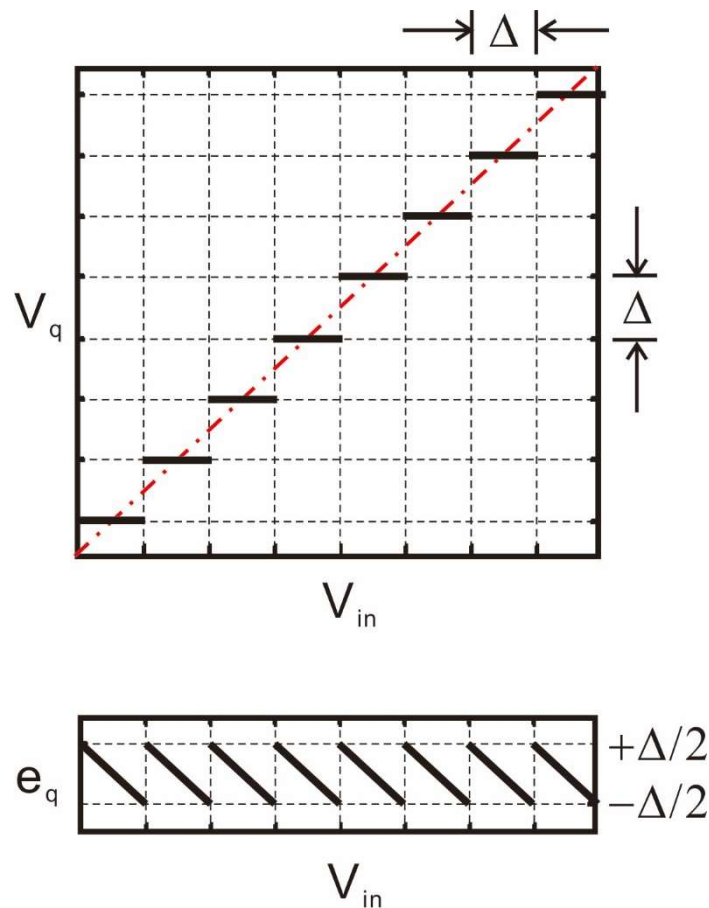


- Aliasing is exploited to down-convert a high-frequency signal.
- High-Q band-pass filters are needed
 - They are hard to design.
- Noise folding is also a severe problem.

Analog-to-Digital Conversion

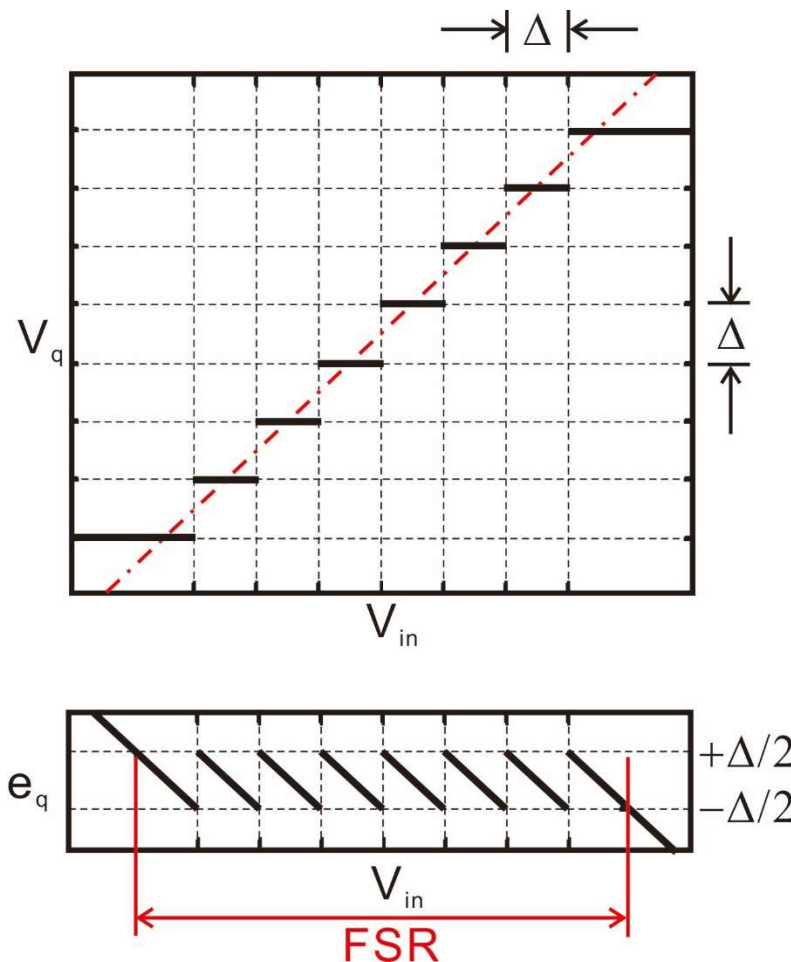


Quantization of an Analog Signal



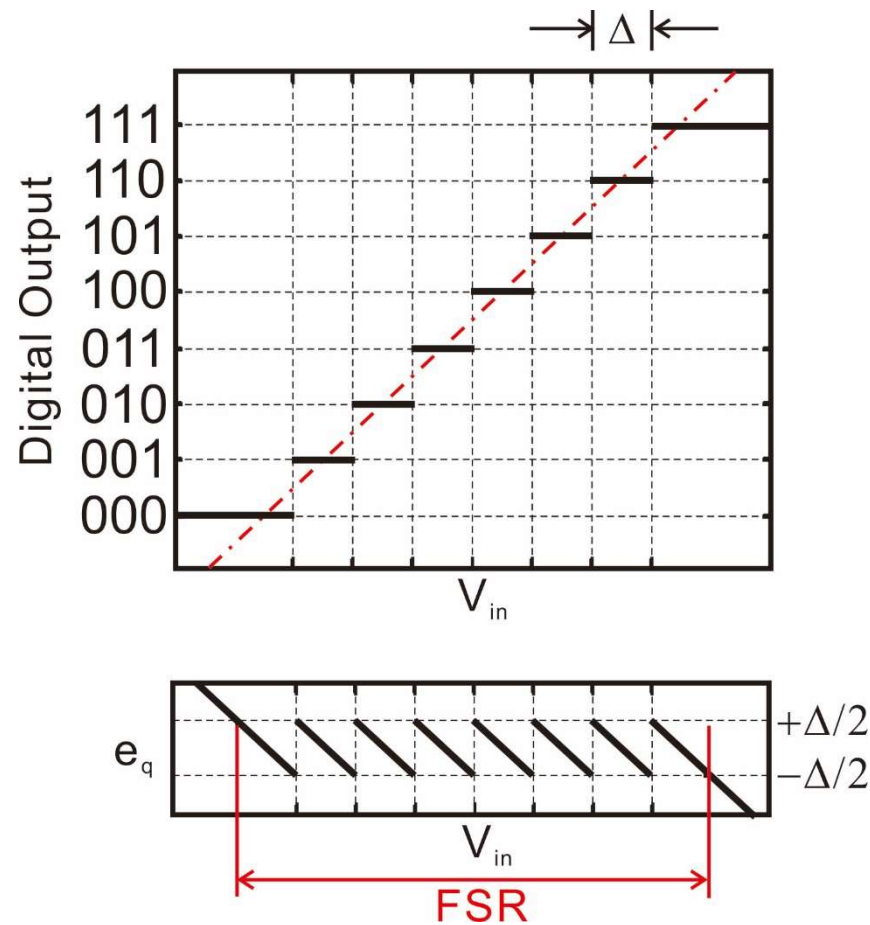
- V_q : quantized voltage
- e_q : quantization error
- $e_q = V_q - V_{in}$
- Quantization step: Δ
- Quantization error has saw tooth shape
 - Bounded by $[-\Delta/2, +\Delta/2]$
- Ideally
 - Infinite input range and infinite number of quantization levels

Quantization of an Analog Signal



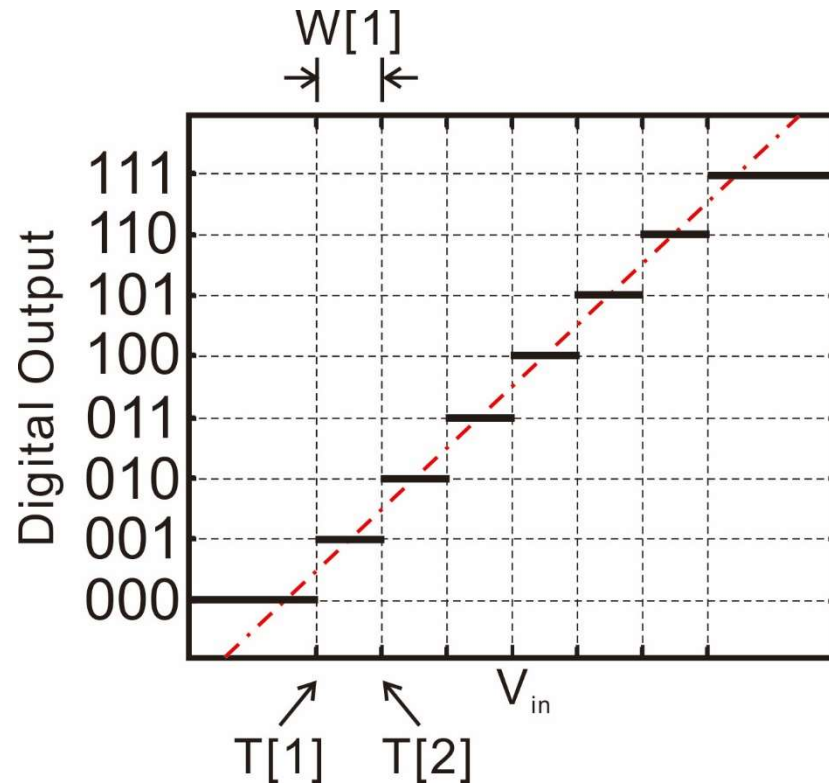
- In practice
 - Finite input range and finite number of quantization levels
- Quantization error grows out of bounds beyond code boundaries
- We define the **full scale range** (FSR) as the maximum input range that satisfies $|e_q| \leq \Delta/2$
- **Overloading** occurs when an input outside the FSR is applied

Quantization of an Analog Signal



- In practice
 - Output is a digital word (not an analog voltage)
- Example: $b=3$
 - $2^3=8$ distinct output codes
 - Other encoding scheme:
 - See e.g. Analog Devices "MT-009: Data Converter Codes"

Nomenclature

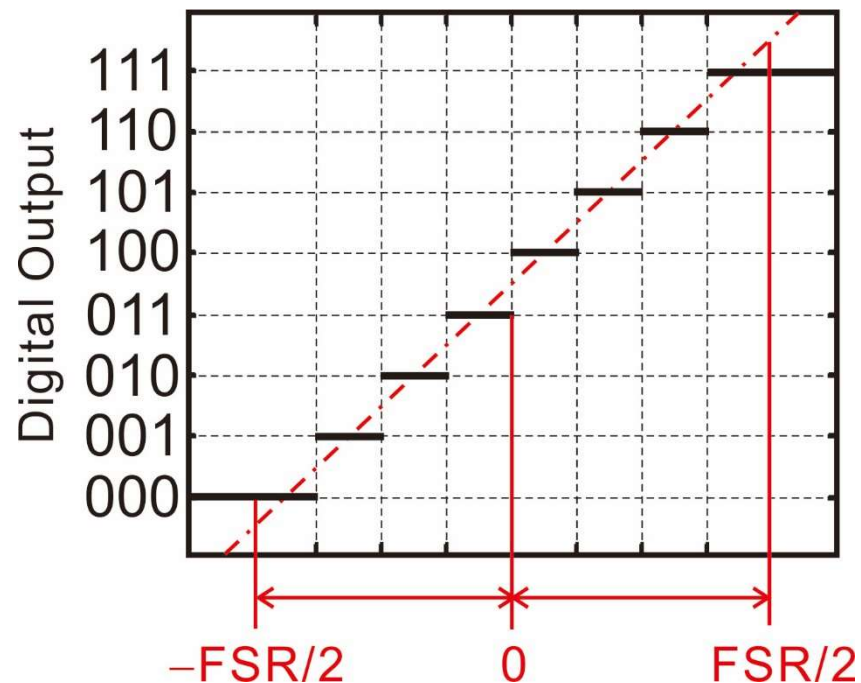


- **LSB size (or width)** – synonymous with code width Δ
- **Transition level** – Input value at the transition between two codes. By standard convention, the transition level $T[k]$ lies between codes $k-1$ and k
- **Code width** – The difference between adjacent transition levels. By standard convention, the code width $W[k] = T[k+1] - T[k]$

Quantizer Classification

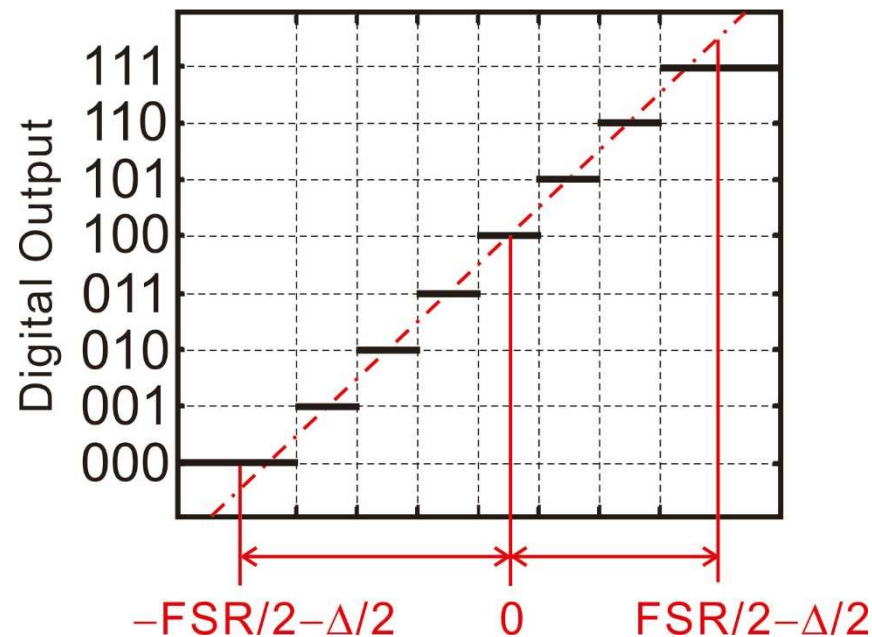
- Based on the zero input location, the quantizers can be classified into:
 - Bipolar input
 - Mid-rise
 - Mid-tread
 - Unipolar input

Bipolar Mid-Rise Quantizer



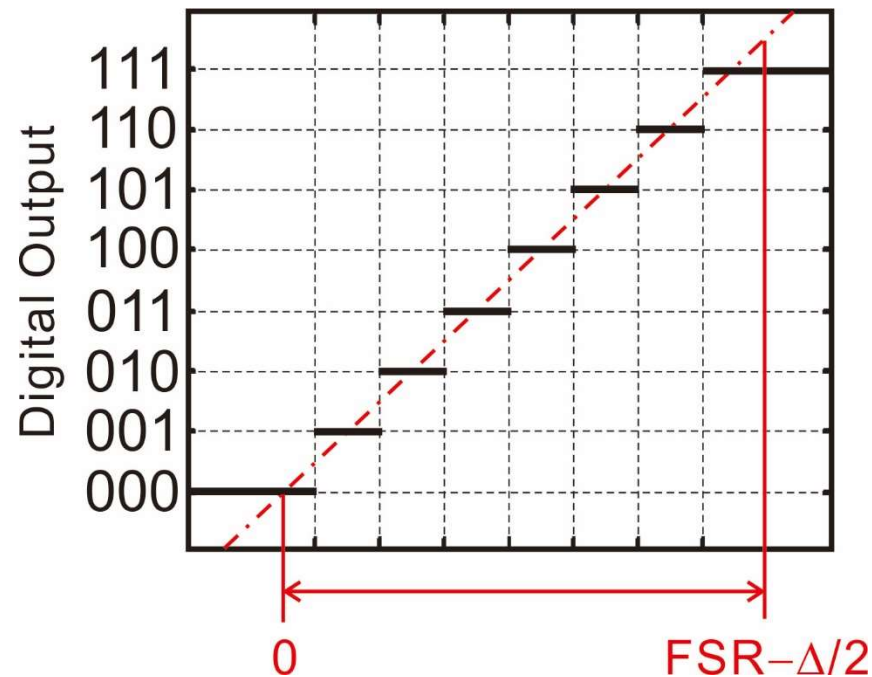
- Bipolar input
 - Differential circuit
- Mid-rise characteristic
 - The center of the transfer function (zero), coincides with a transition level

Bipolar Mid-Tread Quantizer



- Insensitive to noise at 0 input
- In practice, offsets larger than $\Delta/2$ (due to device mismatch) often make this argument irrelevant

Unipolar Quantizer



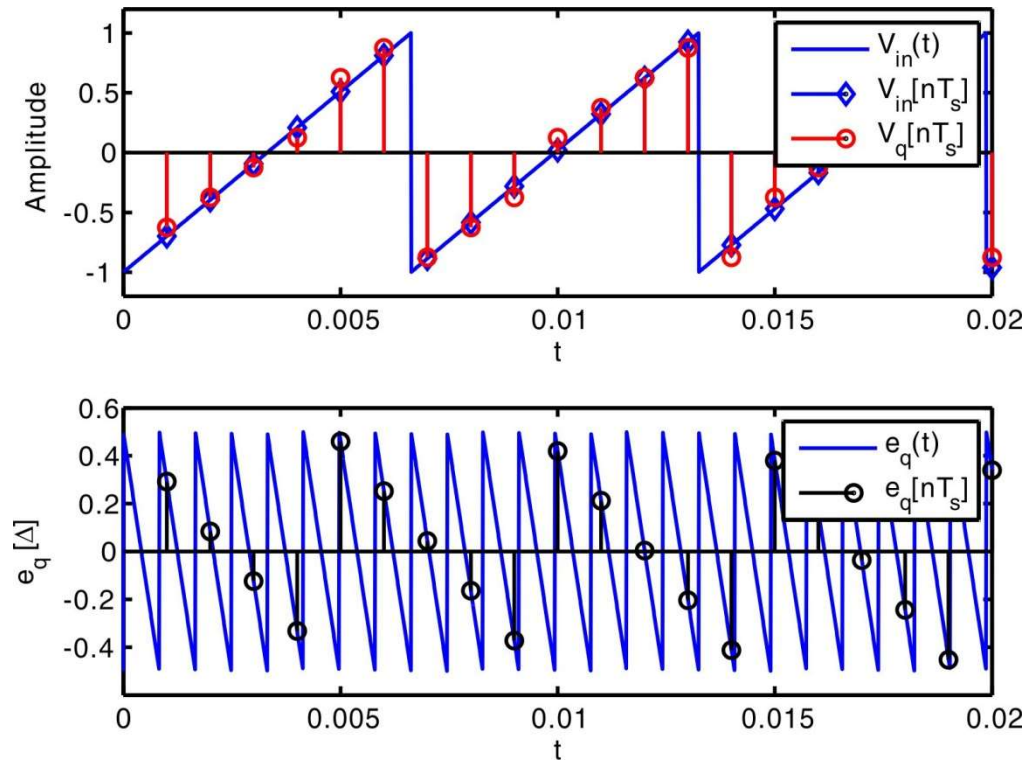
- Usually define origin where first code and straight line fit intersect
 - Otherwise, there would be a systematic offset
- Usable range is reduced by $\Delta/2$

Questions to ask

- How large is the power of the quantization error?
 - Does it depend on the signal?
- How is it distributed in frequency?
 - Is it white?
- Let us consider several cases
 - Ramp input
 - Gaussian random input
 - Sinusoidal input

Linear Ramp Input

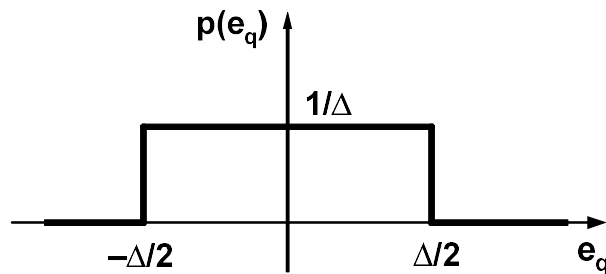
- 8-level quantizer, full-swing input, $f_{in} = 101$ Hz, $f_s = 1000$ Hz



- What is the average power of e_q ?
- Key observation: e_q is uniformly distributed in $[-\Delta/2, \Delta/2]$

Statistical Model of Quantization Error

- e_q is uniformly distributed in $[-\Delta/2, \Delta/2]$



Mean

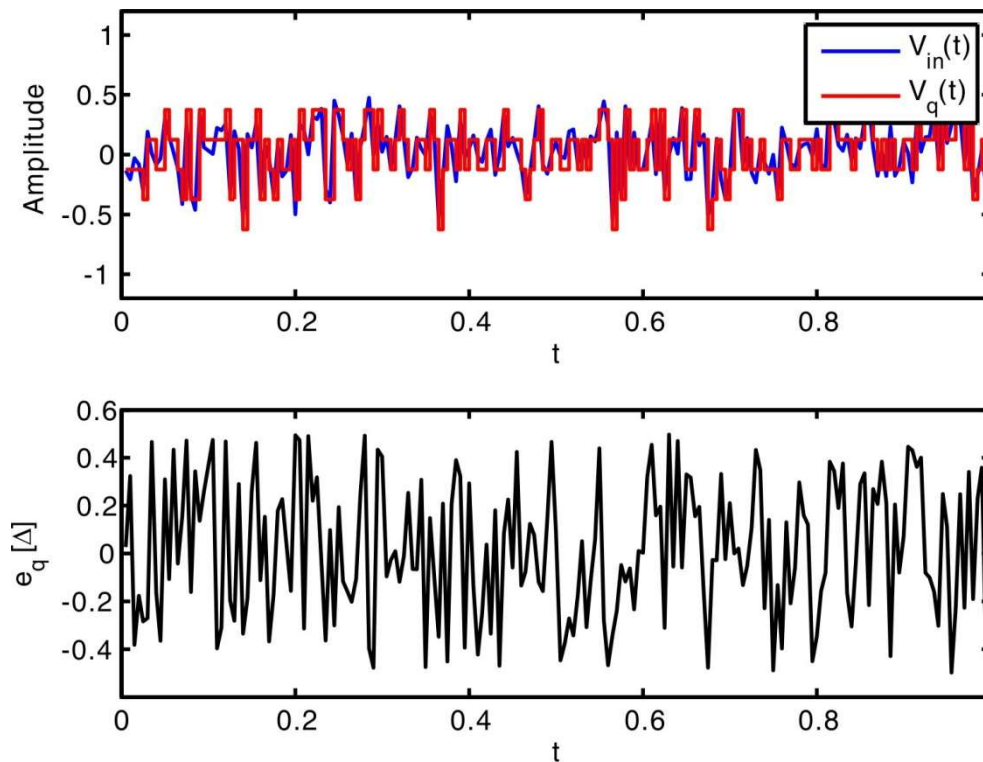
$$\mu = \int_{-\Delta/2}^{\Delta/2} \frac{e_q}{\Delta} de_q = 0$$

Variance

$$\sigma^2 = \int_{-\Delta/2}^{\Delta/2} \frac{e_q^2}{\Delta} de_q = \frac{\Delta^2}{12}$$

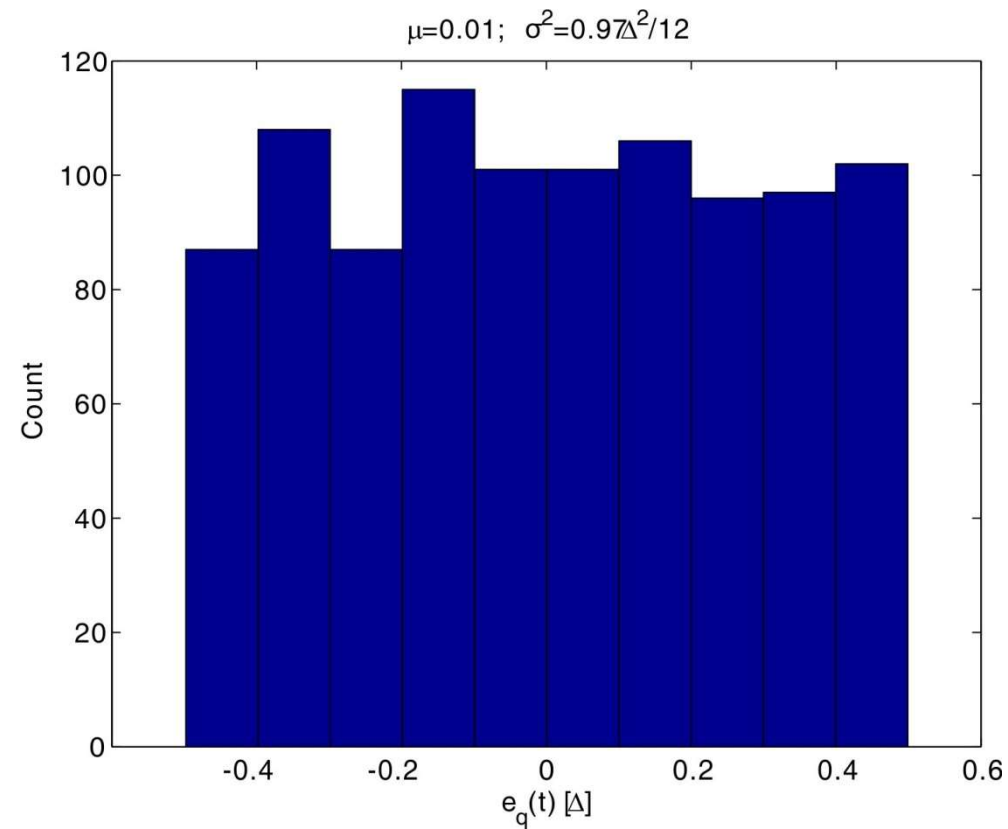
Gaussian Random Input

- 8-level quantizer, V_{in} follows normal distribution with mean of 0 and standard deviation of $1/4$.



- Is e_q uniformly distributed?

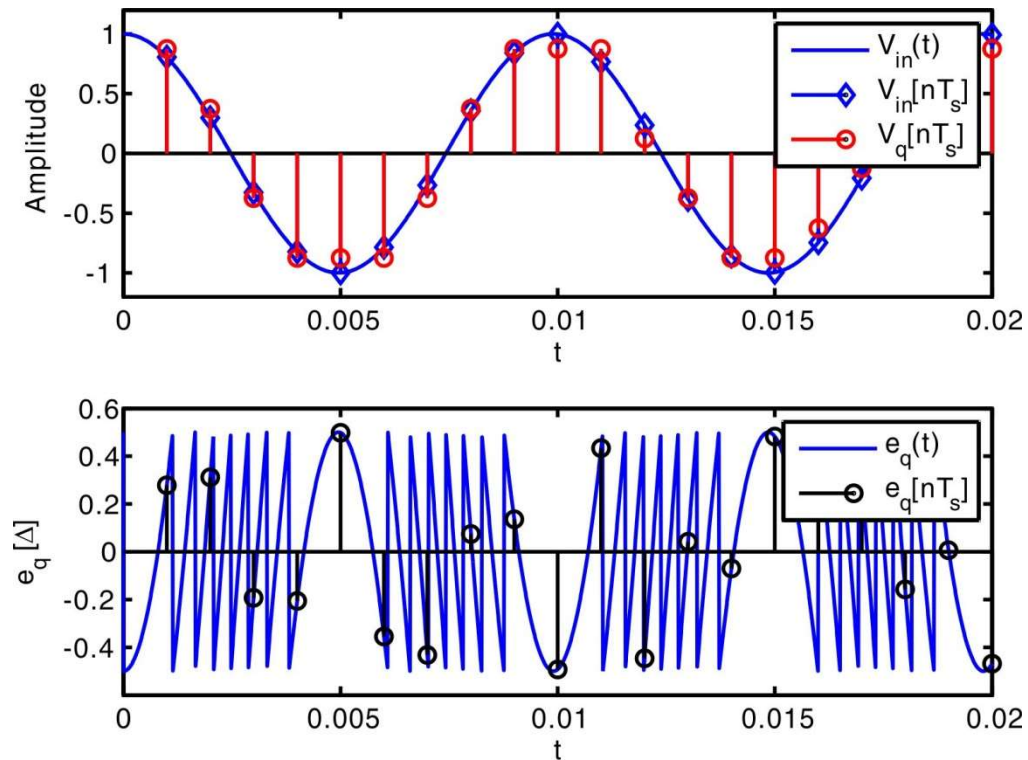
Histogram of Random Input



- Close to uniform distribution

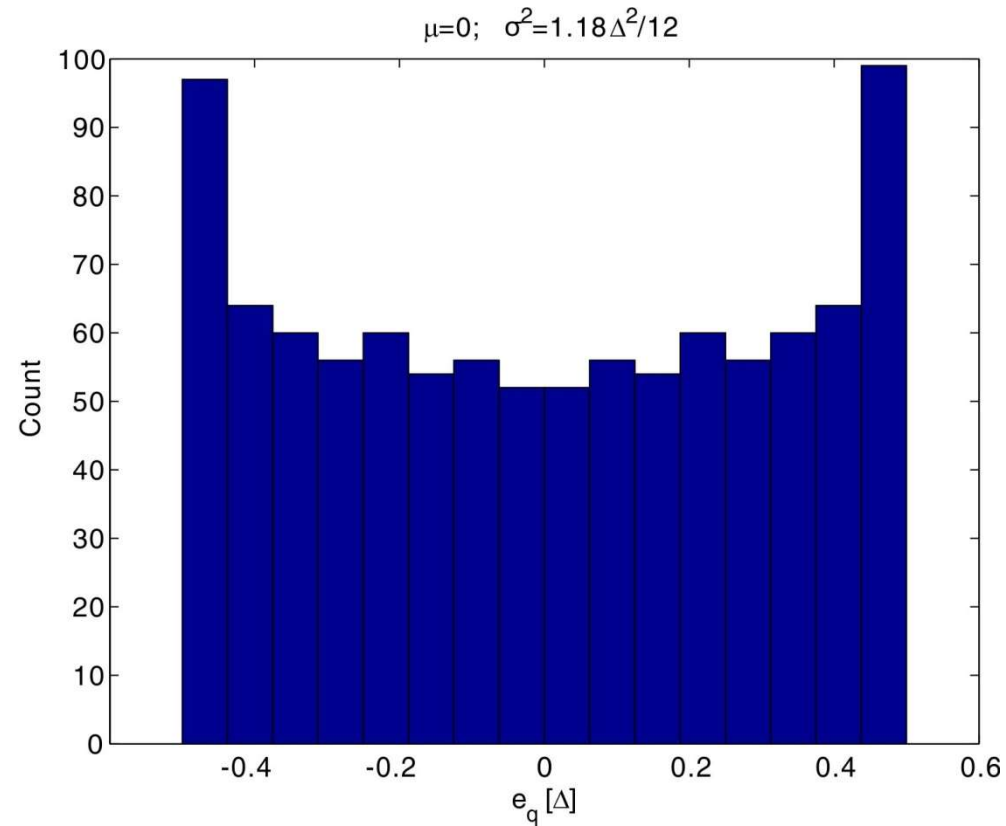
Sine Wave Input

- 8-level quantizer, full-swing input, $f_{in} = 101$, $f_s = 1000$.



- Is e_q uniformly distributed?

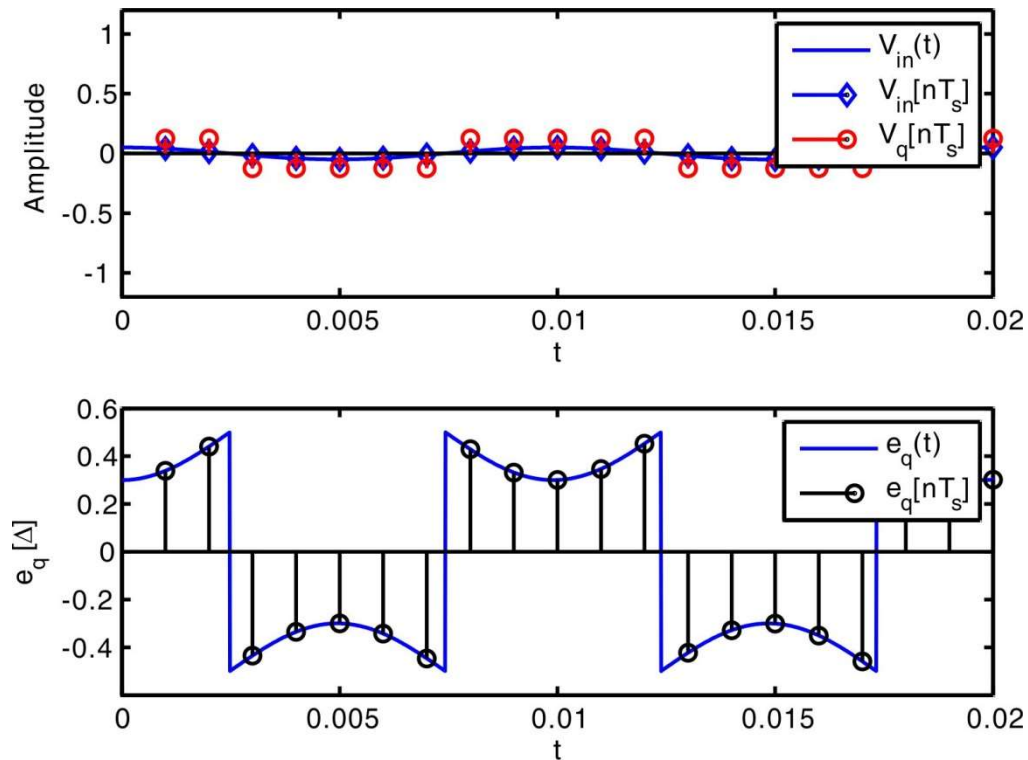
Quantization Error Histogram



- Not exactly uniform distribution, but close
- Does uniform distribution always hold?

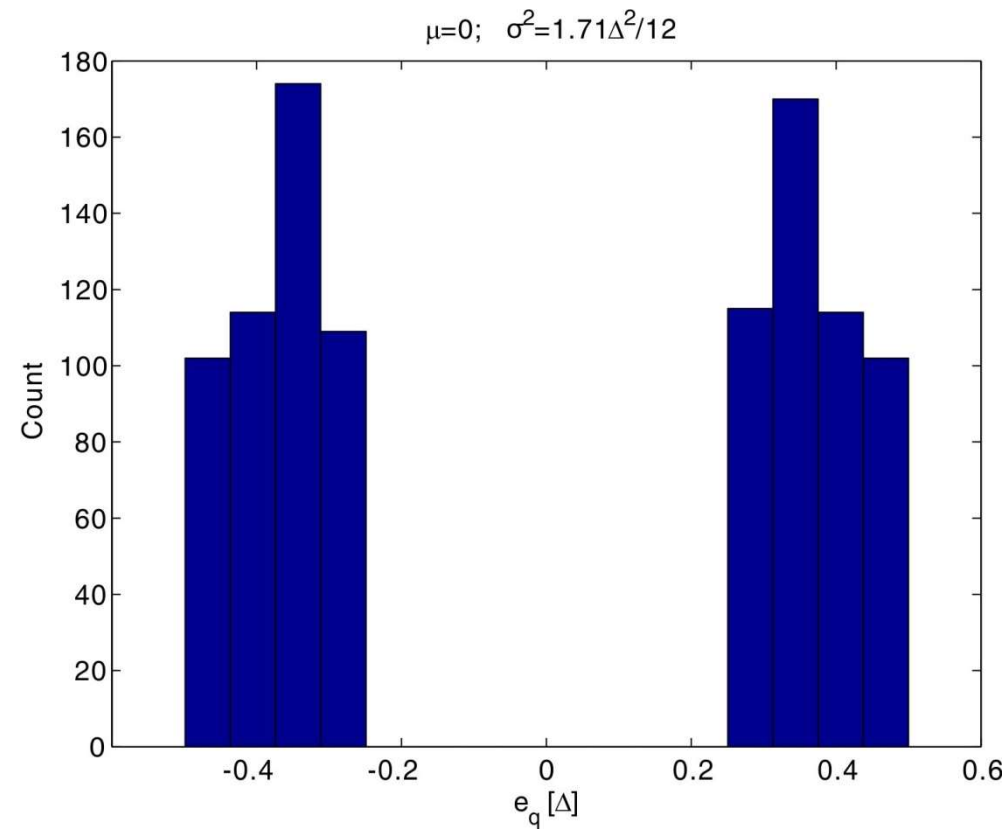
Small Sine Wave Input

- $A = 0.05$;



- Is e_q uniformly distributed?

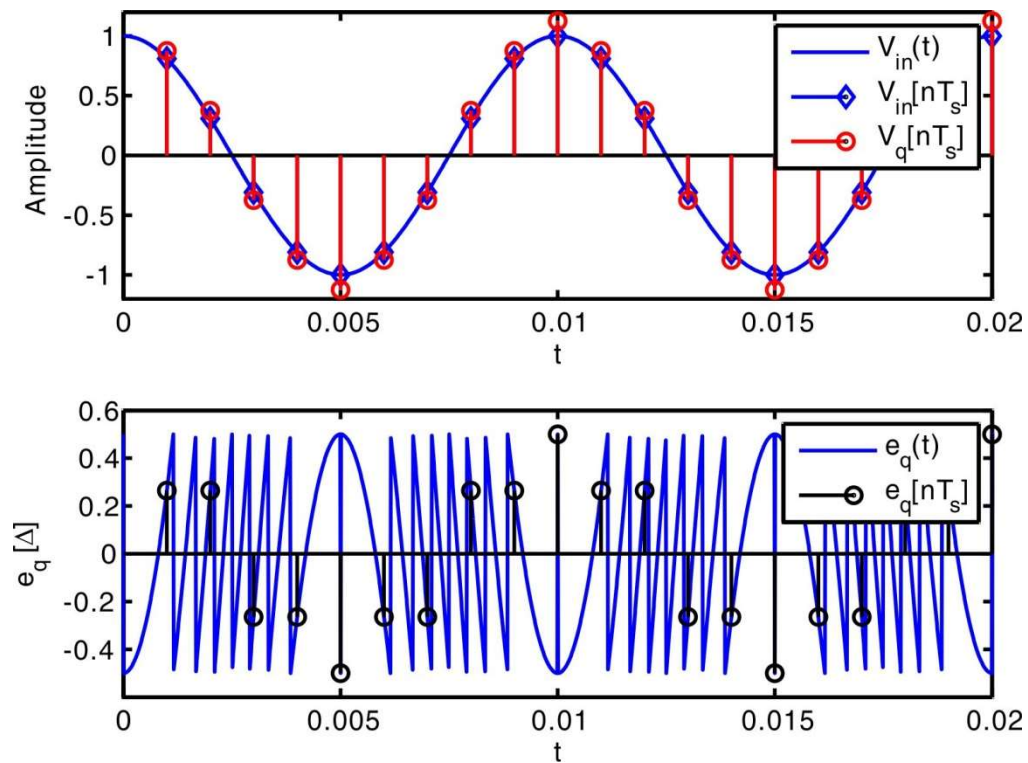
Quantization Error Histogram



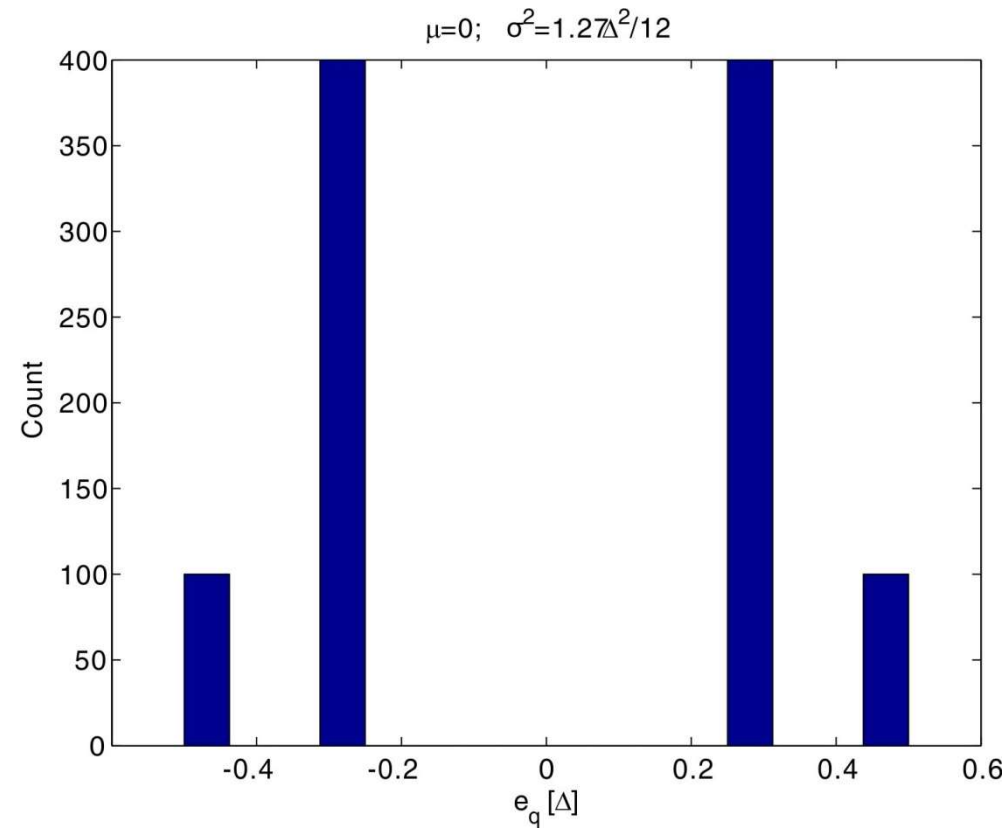
- Why not?

Full Swing Input with $f_{in}=f_s/10$

- Another sine wave example, but now $f_{in}/f_s=100/1000$



Quantization Error Histogram



- e_q has only a few different values!

Explanation

$$V_{in}(n) = \cos(2\pi \frac{f_{in}}{f_s} n) = \cos(\frac{2\pi n}{10}) = \cos(\frac{2\pi(n + 10m)}{10})$$

- Signal repeats every 10 samples, and thus, e_q also repeats
- If we want more distinctive samples and a longer repetition time, we should make f_{in} and f_s mutually prime (or even irrational f_{in}/f_s)

Summary

- Uniform distribution in general holds reasonably well, especially for random input
- However, it does not hold when:
 - The input is too small or DC that it spans only 1 or 2 quantization steps
 - The input is highly correlated with the sampling clock
 - The input is too large that it overloads the quantizer

Signal-to-Quantization-Noise Ratio

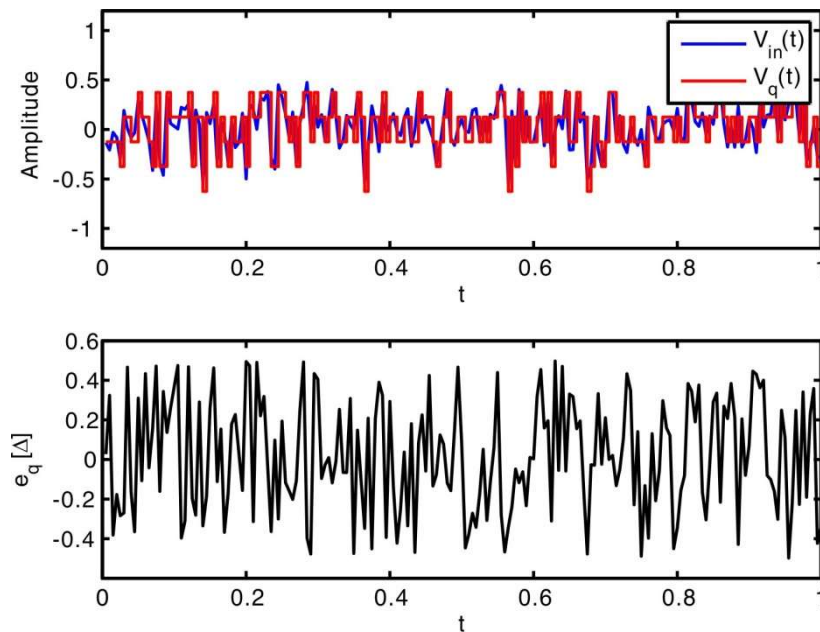
- Signal-to-quantization-noise ratio (SQNR) for an ADC is usually defined as the maximum sinusoidal signal power over the quantization noise power (assuming e_q uniformly distributed)

$$\text{SQNR} = \frac{P_{\text{sig,max}}}{P_{\text{qnoise}}} = \frac{\frac{1}{2} \left(\frac{2^b \Delta}{2} \right)^2}{\frac{\Delta^2}{12}} = 1.5 \times 2^{2b} \approx (6b + 2) \text{ dB}$$

b (Number of Bits)	SQNR
6	38 dB
10	62 dB
14	86 dB
18	110 dB

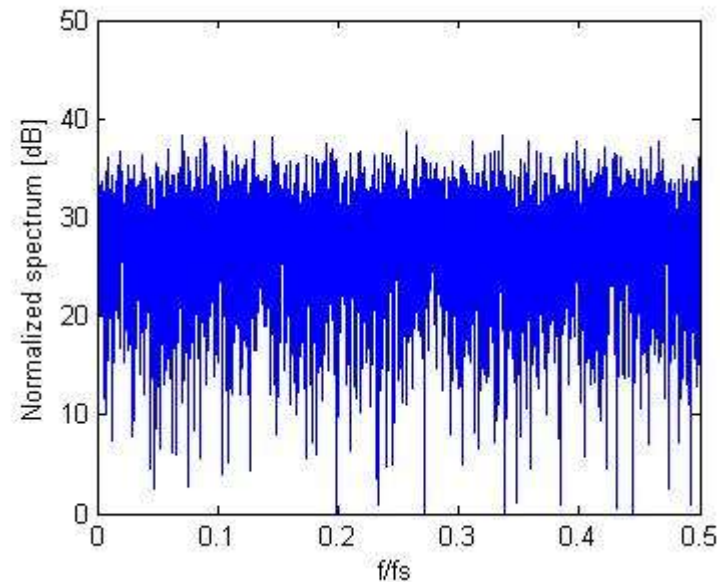
e_q Spectrum for Random Input

- Is the quantization noise spectrum white for a random input?



- e_q looks random.

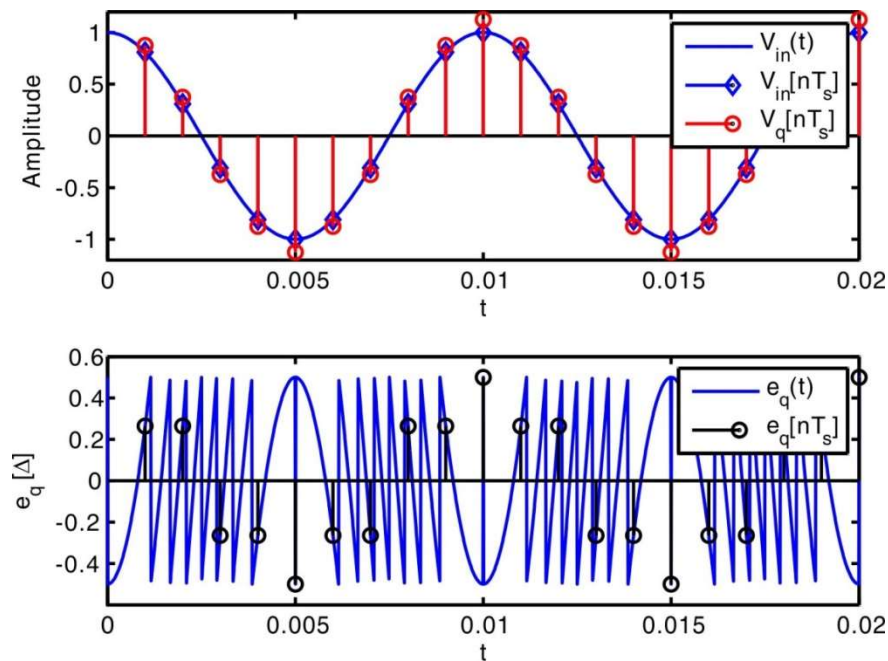
e_q spectrum



- e_q spectrum almost flat.

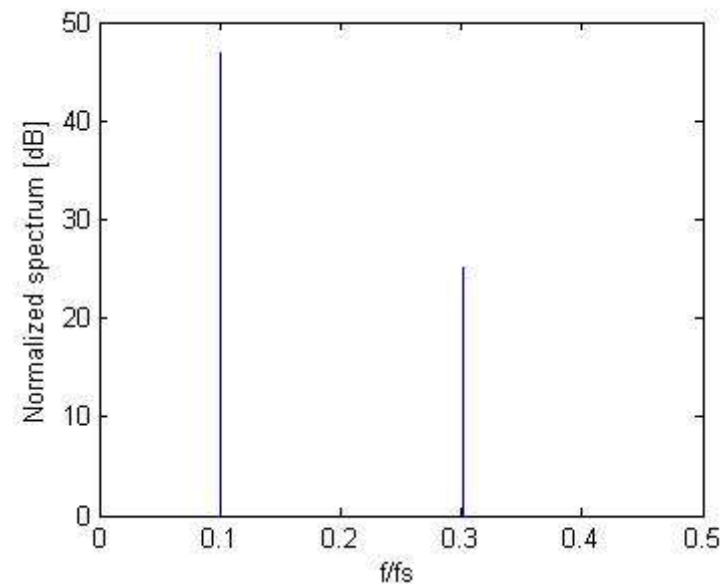
e_q Spectrum for Sine Input

- Is the quantization noise spectrum white?
 - $f_{in}/f_s = 100/1000$, 3-bit quantizer



- e_q is weakly correlated with V_{in} .

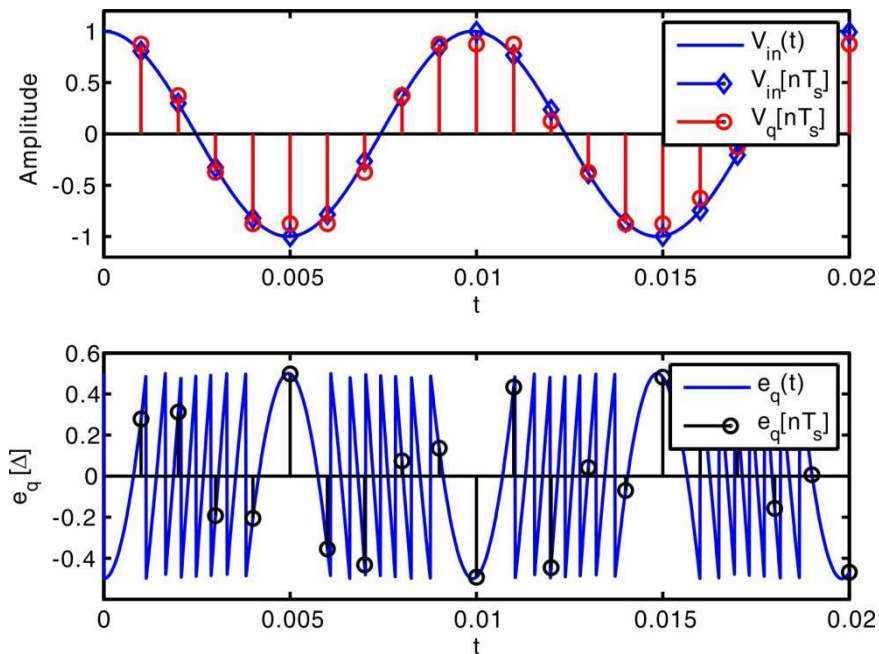
e_q spectrum



- Some harmonics

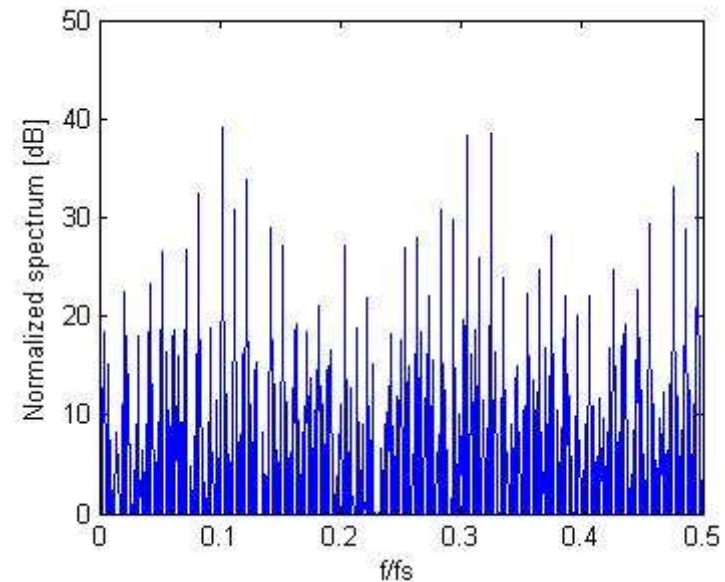
e_q Spectrum for Sine Input

- Is the quantization noise spectrum white?
 - $f_{in}/f_s = 101/1000$, 3-bit quantizer



- e_q is weakly correlated with V_{in} .

e_q spectrum

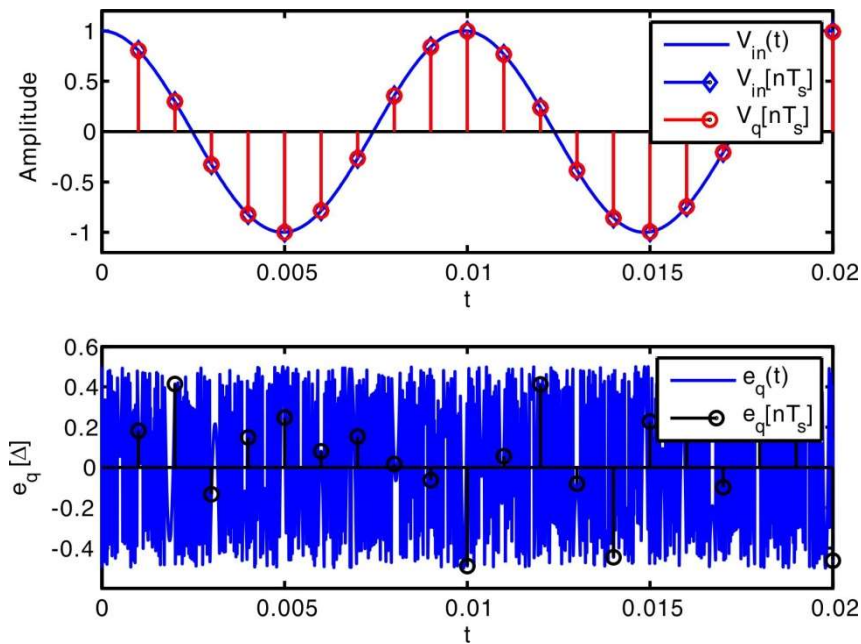


- Some harmonics

e_q Spectrum for Sine Input

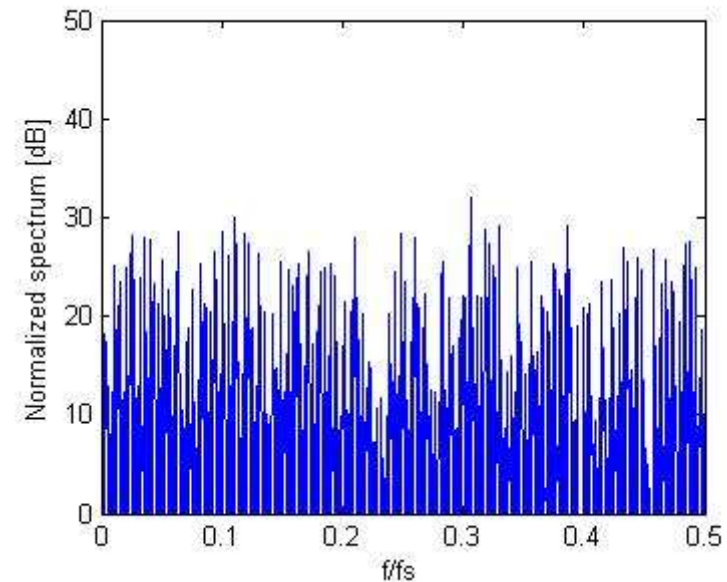
- Is the quantization noise spectrum white?

– $f_{in}/f_s = 101/1000$, 10-bit quantizer



- The correlation becomes weaker

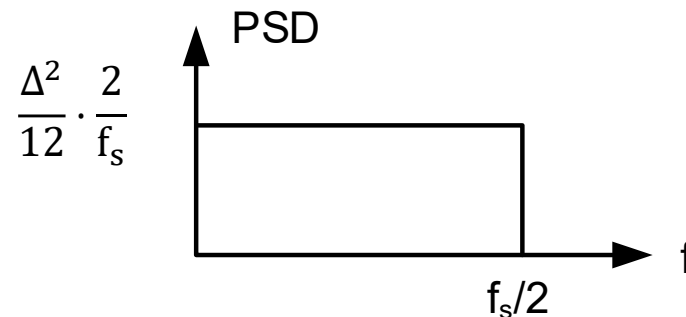
e_q spectrum



- Spectrum almost white

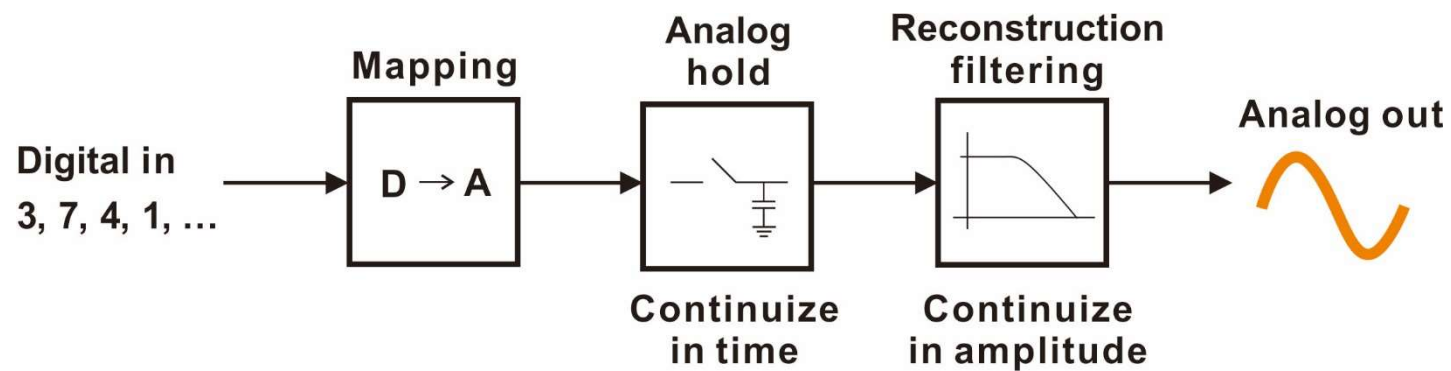
Summary

- For simplicity, we usually assume that the quantization noise power is uniformly distributed in the frequency domain.

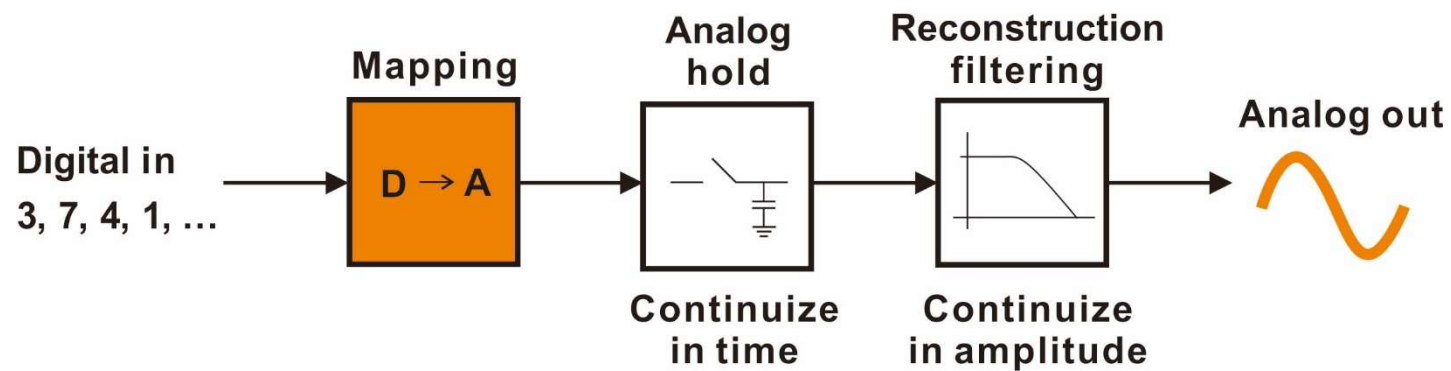


- This approximation holds well for random input (with electronic noise added) and large quantizer resolution.
- More details can be found at:
 - W. R. Bennett, "Spectra of quantized signals," Bell Syst. Tech. J., pp. 446, 1948.
 - B. Widrow, "A study of rough amplitude quantization by means of Nyquist sampling theory," IRE Trans. Circuit Theory, vol. CT-3, pp. 266-76, 1956.
 - A. Sripad and D. A. Snyder, "A necessary and sufficient condition for quantization errors to be uniform and white," IEEE Trans. Acoustics, Speech, and Signal Processing, pp. 442-448, 1977.

Digital-to-Analog Conversion

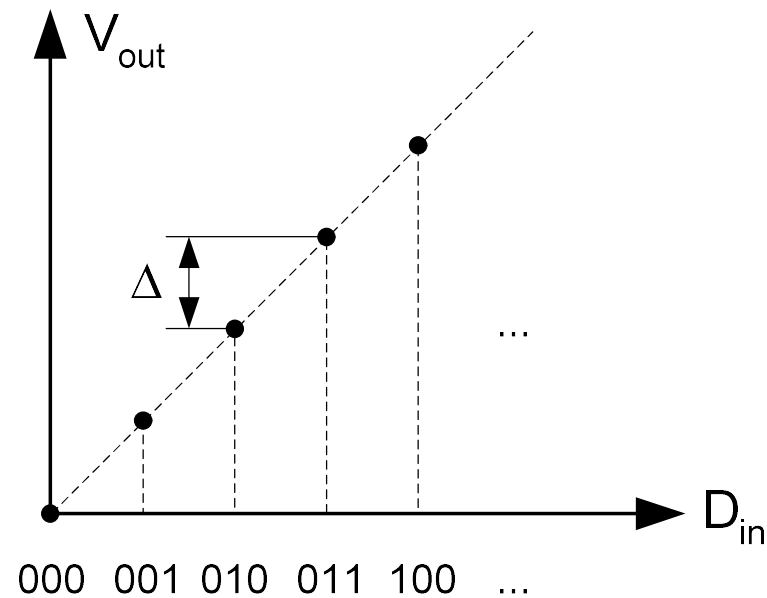


Digital-to-Analog Conversion

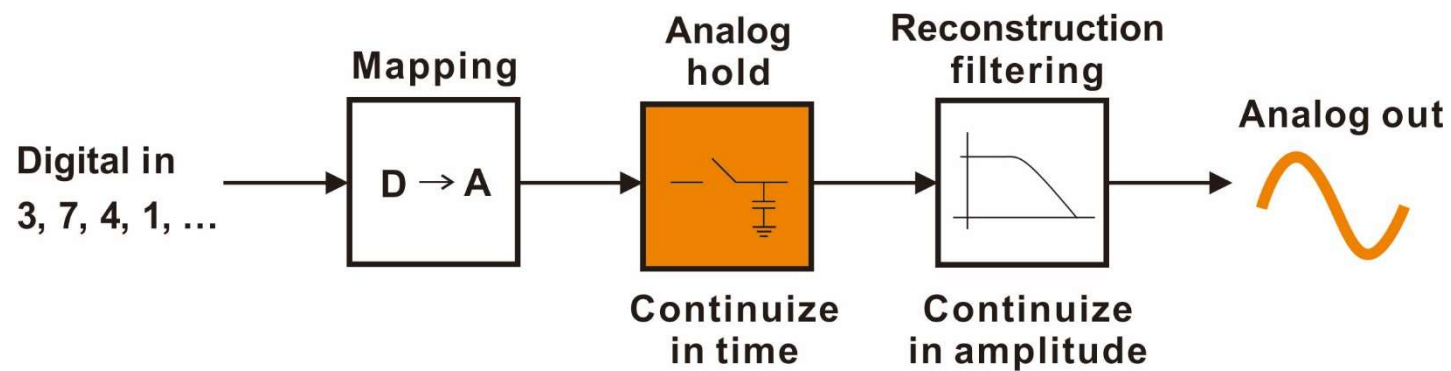


Ideal DAC Mapping

- Map a digital input to an analog quantity
- Is there any quantization error for DAC?



Digital-to-Analog Conversion

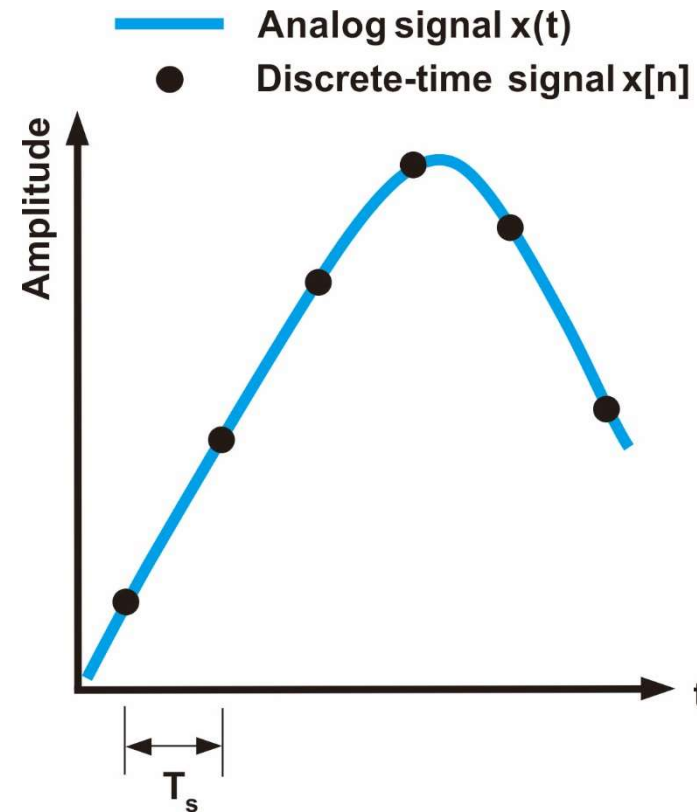


Reconstruction Problem

- How to reconstruct $x(t)$ from $x[n]$?
- Ideal interpolation:

$$x(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin(\pi f_s t)}{\pi f_s t}$$

- However, impossible to implement
 - sinc is noncausal

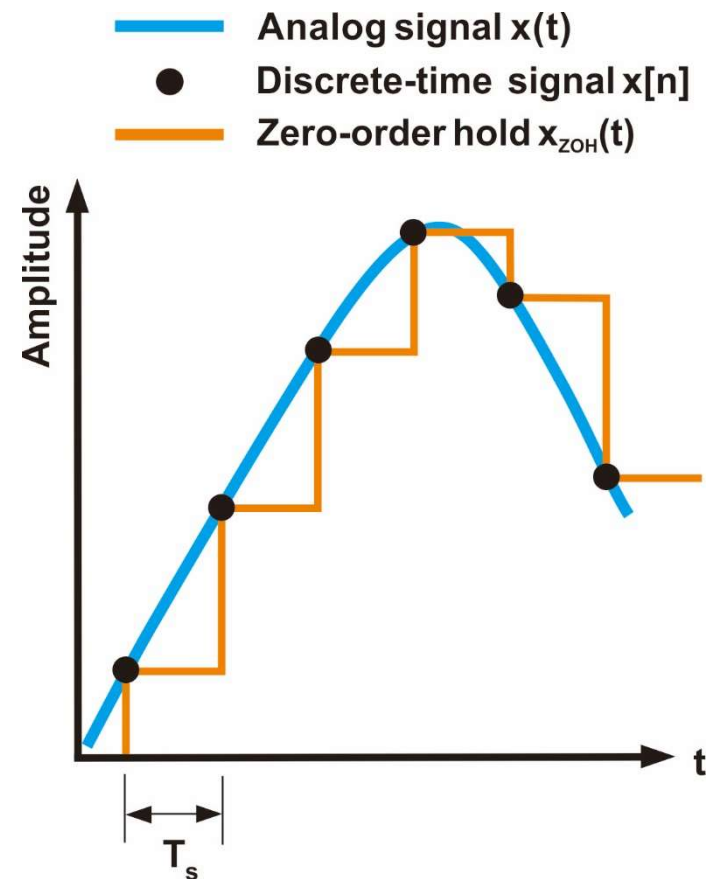


Zero-Order-Hold Reconstruction

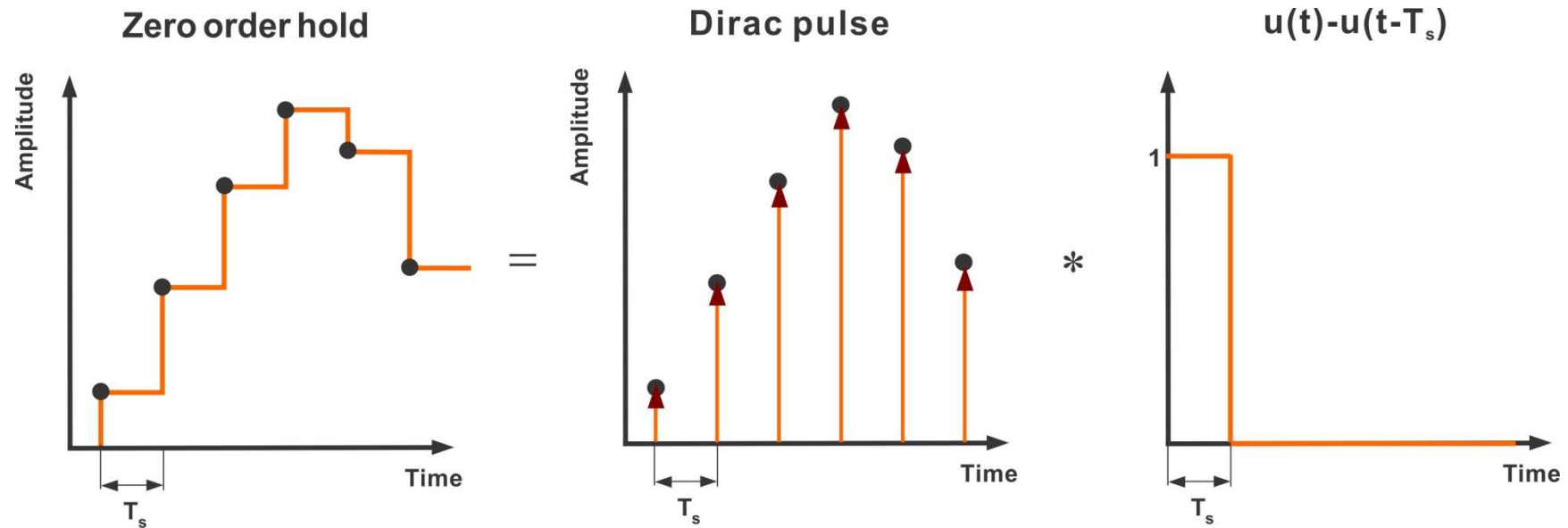
- We can simply hold the discrete time values
- What does this do to the signal spectrum?

$$x_{\text{ZOH}}(t) = x_d(t) * (u(t) - u(t - T_s))$$

$$x_d(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$



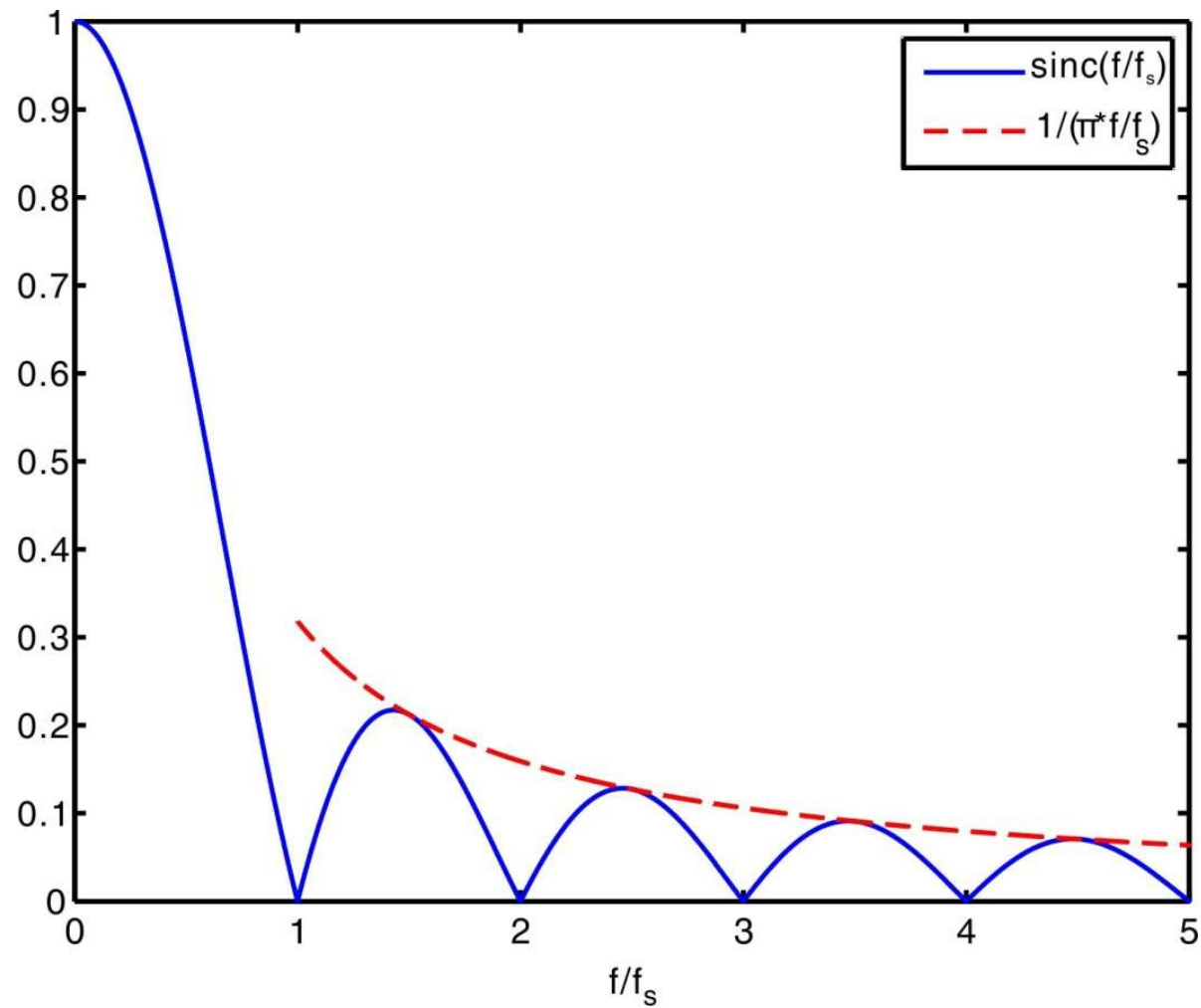
$$x_{\text{ZOH}}(t) = x_d(t) * (u(t) - u(t - T_s))$$

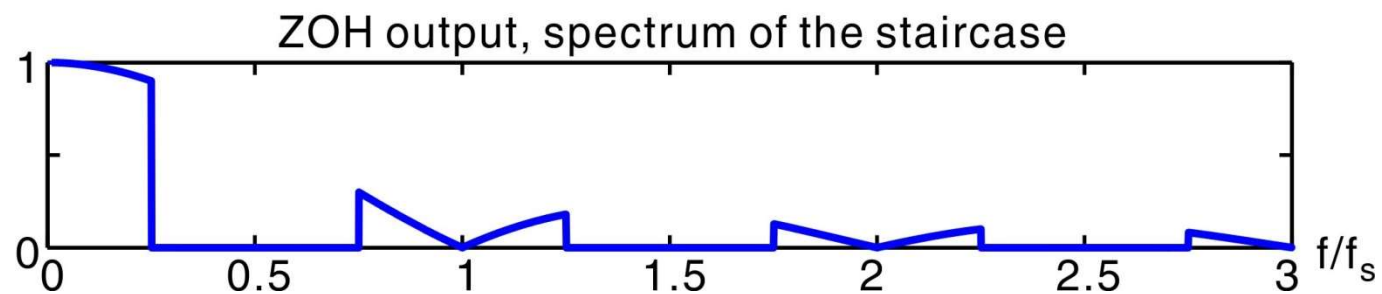
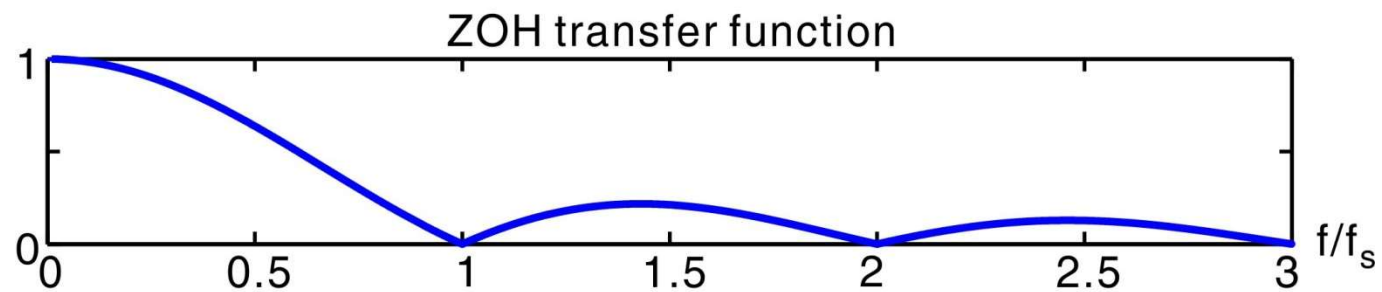
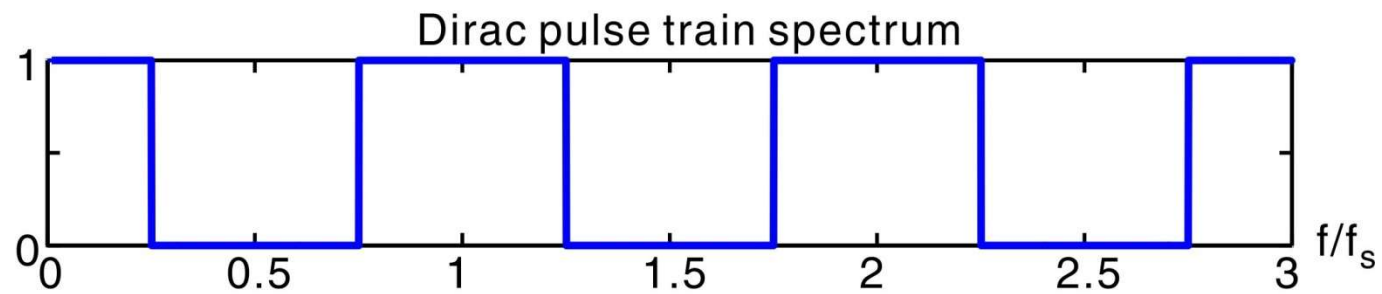
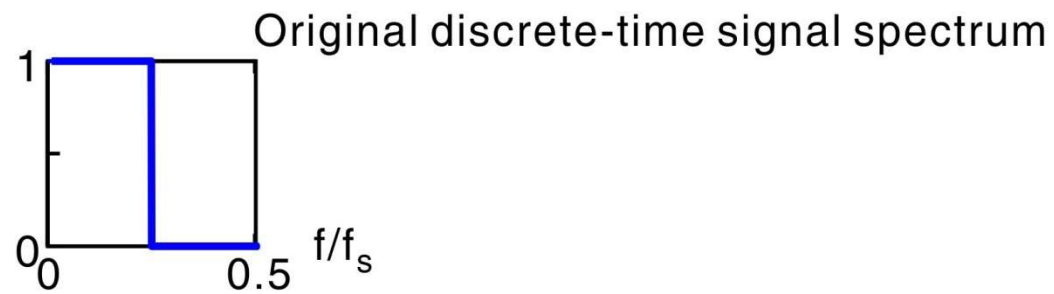


$$\begin{aligned}
 X_{\text{ZOH}}(f) &= X_d(f) \times H_{\text{ZOH}}(f) \\
 &= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s) \times \frac{\sin(\pi f / f_s)}{\pi f} e^{-j\pi f / f_s} \\
 &= \sum_{n=-\infty}^{\infty} X(f - nf_s) \times \underbrace{\frac{\sin(\pi f / f_s)}{\pi f / f_s}}_{\text{Magnitude}} \underbrace{e^{-j\pi f / f_s}}_{\text{Delay}}
 \end{aligned}$$

Envelope Attenuation of ZOH

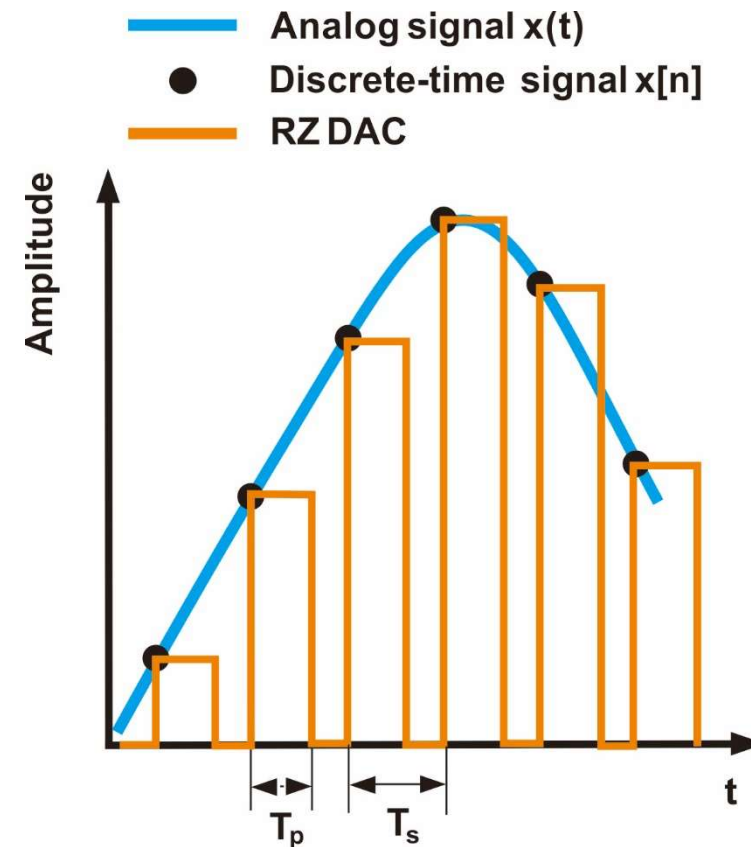
$$\text{sinc}(f/f_s) = \frac{\sin(\pi f/f_s)}{\pi f/f_s}$$

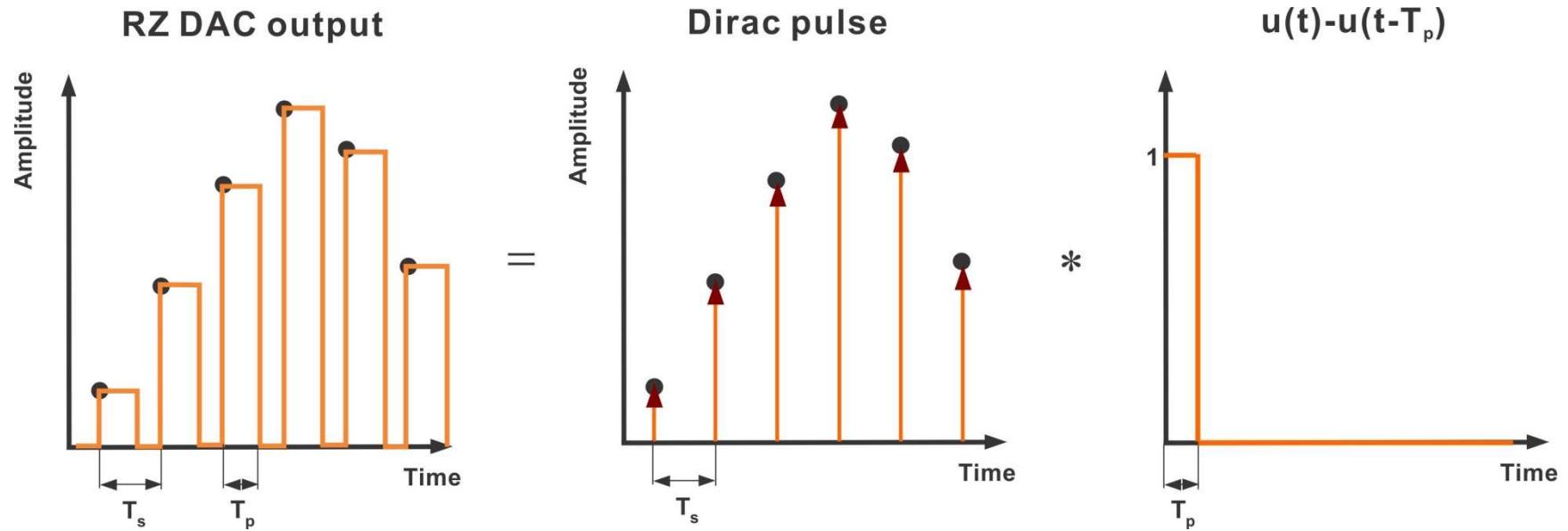




Finite Hold Pulse (RZ DAC)

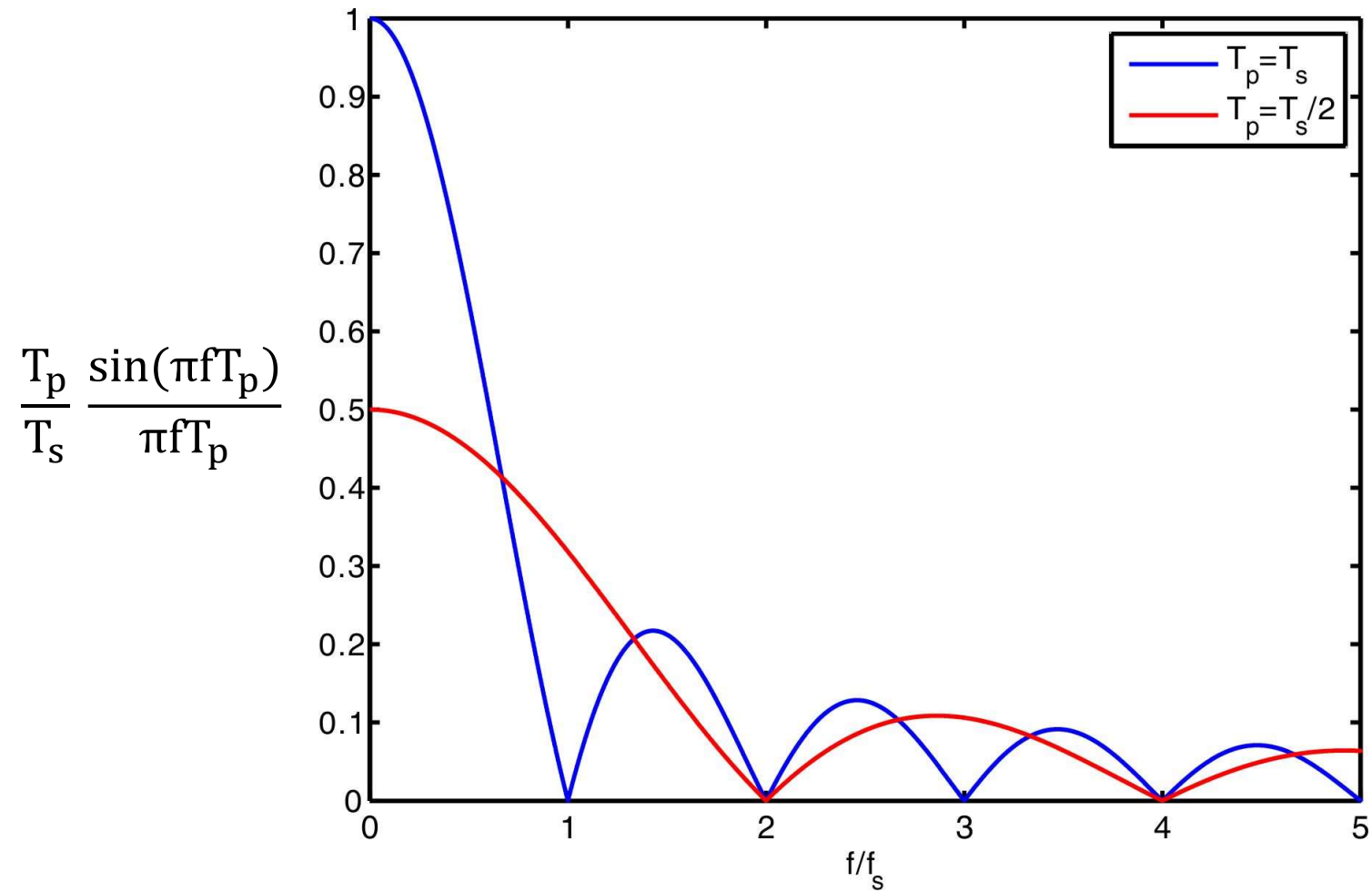
- Hold output for $T_p \leq T_s$
- This is equivalent to convolving the Dirac sequence with a rectangular pulse with reduced width of T_p .





$$\begin{aligned}
 X_{\text{ZOH}}(f) &= X_d(f) \times H_{\text{Hold}}(f) \\
 &= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s) \times \frac{\sin(\pi f T_p)}{\pi f} e^{-j\pi f T_p} \\
 &= \sum_{n=-\infty}^{\infty} X(f - nf_s) \times \underbrace{\frac{T_p}{T_s} \frac{\sin(\pi f T_p)}{\pi f T_p}}_{\text{Magnitude}} \underbrace{e^{-j\pi f T_p}}_{\text{Delay}}
 \end{aligned}$$

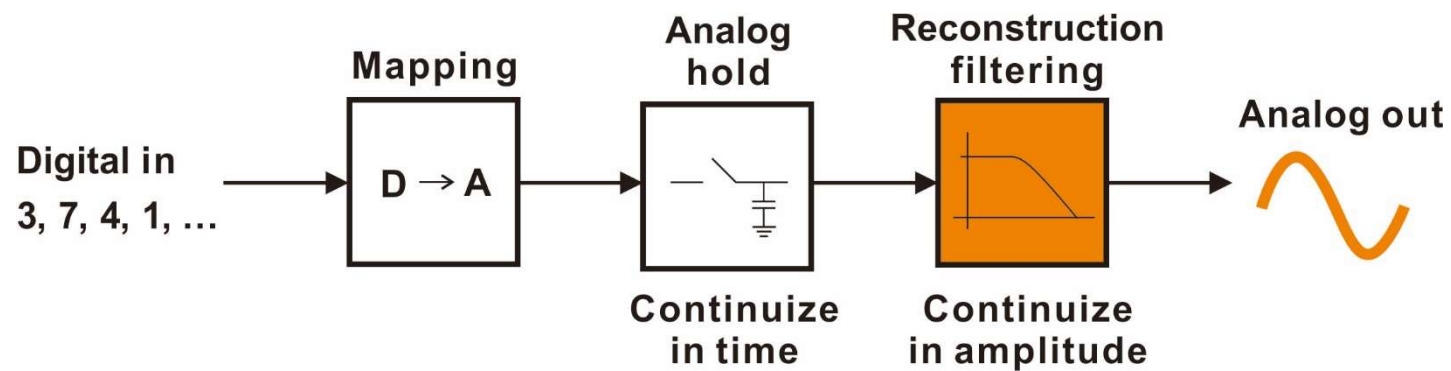
Envelope with $T_p = T_s/2$



NRZ vs. RZ DACs

- Advantage of NRZ DACs
 - Less high-frequency noise → less burden on the filter
 - Less sensitivity to clock jitter
 - Less requirement on slew rate
 - Less power (smaller unit current source)
- Disadvantage of NRZ DACs
 - Large inter-symbol interference
 - Unmatched rise and fall

Digital-to-Analog Conversion



Reconstruction Filter

- Attenuation can be enhanced by increasing:
 - Oversampling ratio
 - Filter order
- We can add pre/post-emphasis to cancel the droop due to the sinc envelope

